

Probability-current analysis of competing pathways for polaritonic chemistry in general harmonic environments

Jonas Vinther,^{1,2} Leonardo A. Cunha,^{3,4} and Johannes Flick^{3,5,6}

¹⁾*Department of Computer Science, University of Copenhagen, DK^{a)}*

²⁾*Niels Bohr Institute, University of Copenhagen, DK*

³⁾*Center for Computational Quantum Physics, Flatiron Institute, 162 5th Avenue, New York, New York 10010, USA*

⁴⁾*Department of Chemistry and Biochemistry, Bates College, Lewiston, Maine 04240, USA*

⁵⁾*Department of Physics, City College of New York, New York, New York 10031, USA*

⁶⁾*Department of Physics, The Graduate Center, City University of New York, New York, New York 10016, USA*

Recent experiments suggest that strong light–matter coupling between molecules and cavity modes can significantly modify chemical reaction rates and yields. Despite multiple proposed mechanisms, a clear microscopic understanding of how vibrational strong coupling alters reaction kinetics remains incomplete. Here, we employ a general quantum-dynamical framework for analyzing competing reaction pathways in polaritonic chemistry by considering reaction coordinates with multiple relevant transition frequencies and general harmonic environments. We introduce a concise formulation for effective spectral densities and a rigorous compression of Markovian bath modes into system-level jump operators. We extend on a tensor-network formulation of the hierarchical equations of motion (HEOM) method to compute transition rates between vibrational states. Decomposing the resulting state-to-state probability currents reveals how cavity-induced changes in individual transitions propagate through multistep reaction networks. We find that these local modifications can alter kinetic bottlenecks and selectively redistribute flux between competing pathways, changing branching ratios under irreversible conditions without substantially affecting the equilibrium yield, in the reversible regime. Finally, for coupled reaction-coordinate, spectator-mode, and cavity environments, we find structured, asymmetric rate profiles arising from interference between hybridized decay channels. Together, these results provide a mechanistic framework for understanding pathway-specific chemical reactivity under vibrational strong coupling.

I. INTRODUCTION

Hybrid light-matter states, known as polaritons, have received increased attention partly due to the control that the light-field brings in tuning to a regime where the joint system exhibits novel phenomena. In polaritonic chemistry^{1–3}, for instance, an ensemble of molecules are strongly coupled to the harmonic modes of a cavity. The vacuum fluctuations alone have been observed to alter the reaction rate and chemical yield of the molecular processes involved. The seminal work from the group of Ebbesen in 2012⁴, provided the first experimental demonstration that hybrid light-matter states could actively modify chemical landscapes. A significant expansion of experimental exploration occurred in 2016 with the demonstration of vibrational strong coupling (VSC)^{5–10}. By strongly coupling the vibrational modes of an ensemble of molecules to the vacuum electromagnetic field, a large modification of the Ground-State chemical reactivity was observed¹¹. Since then, experiments have begun to actively leverage the collective and delocalized properties of polaritons to design new chemical functionalities and achieve precise synthetic control.^{12–18} A detailed account of these and other newly designed chemical functionalities can be found in a recent review¹⁹.

Nonetheless, predicting what reactions can be affected and the extent of the modification requires an under-

standing that has not yet been fully developed. The aim of this paper is to expand the current understanding through model development and clarifying numerical studies.

Rate theories that are fundamentally concerned with the energy barrier between reactant and product populations, such as transition state theory, Grote–Hynes theory or Pollak–Grabert–Hänggi theory, do not capture the features of the polaritonic state^{20,21}. Because these are classical and semi-classical methods, they predict rate modification profiles that are significantly broader than experimental observations and often peak at incorrect frequencies^{22–24}.

Recent quantum dynamical treatments have demonstrated that the sharp resonance in VSC-modified reaction rates can be recovered when the molecular, cavity, and environmental degrees of freedom are treated quantum mechanically^{20,22,25–29}. Complementary theoretical analyses based on effective environmental descriptions and Fermi’s Golden Rule have provided microscopic interpretations of this resonance and its dependence on cavity properties^{20,25–27}, while recent tensor-network approaches have begun to address the additional complexity of multimode and collective systems^{28–30}. These results underscore the importance of quantum dynamics for describing VSC-induced rate modifications and motivate our approach of performing an *exact* quantum dynamical simulation of the reaction coordinate and its environment.

Within such models, perturbative assessment of the spectral-overlap with the noise-spectrum have proven to

^{a)}Corresponding author: jonas.vinther@nbi.ku.dk

provide valuable insight^{20,25–27}. We will expand on this insight by providing a more detailed picture of the transition rate between the vibrational states of the reaction coordinate. Both experiments and theoretical studies have demonstrated that cavity-induced modifications of reaction rates are strongest when the cavity frequency is resonant with the vibrational frequency of the reactant or product wells^{6,11,22}. Therefore, this study will illuminate the interdependence on these in more detail, by considering a model with independent well frequencies. This formulation naturally generalizes to multiple wells, allowing a simple starting point for studying competing pathways in polaritonic chemistry.

This paper is organized as follows. Section II formulates the model of the reaction coordinate and its harmonic environment. It also presents the reaction rate, defined in terms of system observables, and the probability currents, which serves as the diagnostic tool for the reaction mechanics. Section III describes our method for performing the numerically exact simulations of the dynamics. The results are presented and discussed in section IV, where we address potential energy surfaces with multiple relevant transition frequencies and competing pathways and harmonic environments with interfering decay channels. Our conclusions are summarized in sec. V.

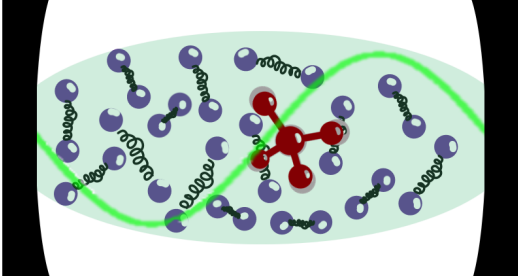


FIG. 1. Visualization of a single molecule (in red) that contains the reaction coordinate and possibly spectating vibrational modes. The molecule is embedded in a solvent, as represented by the phonon bath (purple). The molecule and the solvent are placed in a cavity, which is represented by a single cavity mode (in green).

II. THEORY

A. Model & effective harmonic environment

We employ a complete quantum description of a reaction coordinate (the system) on a one dimensional potential energy surface (PES) in a harmonic environment (the bath). A schematic for the physical setup is presented in Fig. 1.

The microscopic model that we consider, is given by

the Pauli-Fierz Hamiltonian in the length gauge^{22,27,31}:

$$H = H_R + H_Q + H_c + H_R^{\text{CL}} + H_Q^{\text{CL}} + H_c^{\text{CL}} \quad (1)$$

$$H_R = \frac{\hat{P}^2}{2M} + V(\hat{R}) \quad (2)$$

$$H_Q = \frac{\hat{P}_Q^2}{2} + \frac{\omega_Q^2}{2} \left(\hat{Q} - \frac{c_Q}{\omega_Q^2} \hat{R} \right)^2 \quad (3)$$

$$H_c = \frac{\hat{p}_c^2}{2} + \frac{\omega_c^2}{2} \left(\hat{q}_c + \sqrt{\frac{2}{\omega_c}} \eta_c \left(\eta_R \hat{R} + \eta_Q \hat{Q} \right) \right)^2 \quad (4)$$

In terms of positional operators, this Hamiltonian describes a reaction coordinate $\hat{R}$, coupled to a spectator mode $\hat{Q}$, both of which are coupled to a single cavity mode $\hat{q}_c$. Furthermore, each of these coordinates are assumed to experience their own harmonic environment, as described by the Caldeira-Leggett model of dissipation³², via the inclusion of $H_{(R,Q,c)}^{\text{CL}}$ in Eq. 1. These are of the form:

$$H_R^{\text{CL}} = \sum_l \frac{\hat{p}_l^2}{2} + \frac{\omega_l^2}{2} \left(\hat{x}_l - \frac{c_l}{\omega_l^2} \hat{R} \right)^2 \quad (5)$$

The reaction coordinate (RC) is the only anharmonic degree of freedom in the Hamiltonian, Eq. 1, and all the other harmonic degrees of freedom may be referred to as the bath. Resultingly, if we assume that the RC is initially in a product with the bath and that the bath is in a thermal state, the influence of the bath is fully described by the bath correlation function (BCF)^{33,34}:

$$C(t) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} S_\beta(\omega) e^{-i\omega t} = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} \frac{2J(\omega) e^{-i\omega t}}{1 - e^{-\beta\omega}} \quad (6)$$

where $S_\beta(\omega)$ is the noise spectrum, β is the inverse temperature and $J(\omega)$ is the spectral density of the full harmonic environment.

In the SI. sec. B we provide a succinct formula for $J(\omega)$ for arbitrary harmonic environments. It relies on the inversion of a matrix that can be formed straightforwardly from the parameters of the Hamiltonian, thus mitigating the algebraic labor that earlier methods required^{27,35}.

Figure 2 schematically shows the abstraction from the microscopic harmonic environment in question to the full spectral density $J(\omega)$. With our approach, Eq. 1 immediately leads to:

$$J(\omega) = J_R^{(DL)}(\omega) + \text{Im} [\mathbf{c}^T M^{-1}(\omega) \mathbf{c}] \quad (7)$$

$$\mathbf{c} = \begin{pmatrix} \sqrt{2\omega_c^3} \eta_R \eta_c \\ c_Q + 2\omega_c \eta_c^2 \eta_R \eta_Q \end{pmatrix} \quad (8)$$

$$M(\omega) = \begin{pmatrix} \Gamma_c(\omega) & -\sqrt{2\omega_c^3} \eta_Q \eta_c \\ -\sqrt{2\omega_c^3} \eta_Q \eta_c & \Gamma_Q(\omega) - 2\omega_c \eta_c^2 \eta_Q^2 \end{pmatrix} \quad (9)$$

$$\Gamma_{(c,Q)}(\omega) = \omega^2 - \omega_{(c,Q)}^2 - \left(i + \frac{\omega}{\gamma_{(c,Q)}} \right) J_{(c,Q)}^{(DL)}(\omega) \quad (10)$$

For simplicity, we have assumed that the spectral densities of the individual harmonic environments, described

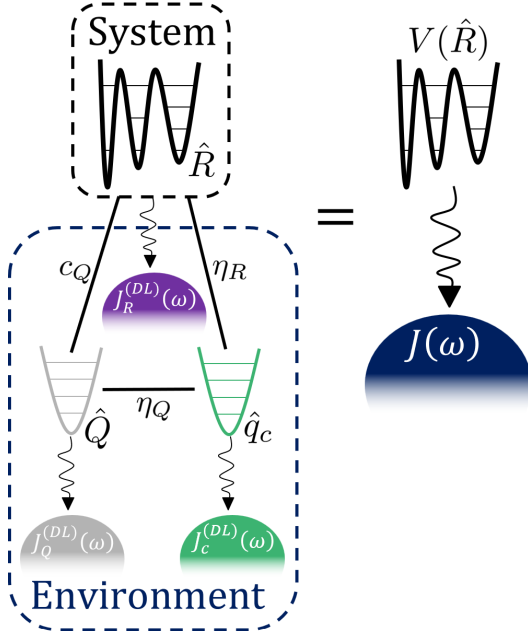


FIG. 2. Schematic for the efficient modeling of the harmonic degrees of freedom in terms of the spectral density for the full harmonic environment. Sec. B in the SI derives and presents the formula for arbitrary harmonic environments. The system refers to the reaction coordinate $\hat{R}$, here visualized in terms of the PES $V(\hat{R})$ for the triple well system treated in sec. IV A 3. The spectator mode has coordinate $\hat{Q}$ and the cavity $\hat{c}$. The couplings between them are denoted by c_Q , η_R and η_Q and $J_{(R,Q,c)}^{(DL)}(\omega)$ refer to the individual baths which we choose to be of the Drude-Lorentz form.

by $H_{(R,Q,c)}^{\text{CL}}$, are all of the Drude-Lorentz form:

$$J_n^{(DL)}(\omega) = \frac{2\lambda_n\gamma_n\omega}{\omega^2 + \gamma_n^2} \quad \text{with } n \in \{R, Q, c\} \quad (11)$$

For the cavity bath, we will furthermore assume Markovianity by considering the limit $\gamma_c \rightarrow \infty$ with $\frac{2\lambda_c}{\gamma_c} = \alpha$ constant³⁶.

In the case of no spectator mode ($c_Q = \eta_Q = 0$), the harmonic environment reduces to that of the phonon bath (modeled by the spectral density $J_R^{(DL)}(\omega)$) plus the lossy cavity²⁰:

$$J(\omega) = J_R^{(DL)}(\omega) + \frac{2\omega_c^3\eta_R^2\eta_c^2\alpha\omega}{(\omega^2 - \omega_c^2)^2 + \alpha^2\omega^2} \quad (12)$$

The resulting noise spectrum is visualized for three different cavity frequencies in figure 3.

As alluded to in figure 2, the cavity-modified kinetics are essentially governed by two scalar function: the potential energy surface $V(\hat{R})$ and the effective spectral density for the full harmonic environment $J(\omega)$. The former accounts for the specifics of the reaction while the latter contains all the harmonic degrees of freedom such as lossy cavity modes, spectator modes and the vibrations of the solvent. Thus, the simple spectral density

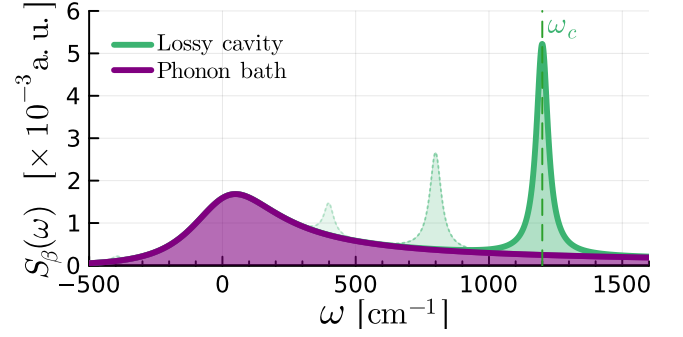


FIG. 3. The noise spectrum when there is no spectator mode ($c_Q = \eta_Q = 0$). The effect of the cavity is captured by the peak centered around ω_c (Eq. 12). The height of this peak scales with $\sim \omega_c^2$, as dictated by the Pauli-Fierz Hamiltonian. For visualizations of noise spectra with the full spectral density of Eq. 7, see Fig. 8

(Eq. 12) will be used to single out the effects of the cavity when studying PES driven effects (sec. IV A). The full spectral density (Eq. 7) will be applied to study how the alignment between the reaction coordinate and spectator mode polarities, encoded in η_R and η_Q respectively, affects the reaction rate profile (sec. IV B).

We parameterize the potential energy surface $V(\hat{R})$ in terms of piecewise parabolic functions, which provides a straightforward way to study independent well frequencies and competing pathways. Explicit expressions for $V(\hat{R})$ are found in the SI. sec. C. While the simulations are carried out in the eigenbasis of the system Hamiltonian H_R , the analysis is performed in a basis where select eigenstates have been hybridized to construct localized states. Our approach for constructing the rotation between these two bases relies on diagonalizing $\hat{R}$ in the subspace of the selected states (see SI. sec. C 3).

B. Observables & Mechanistic Diagnostics

The reaction rate is defined in terms of expectation values of projection operators onto a reactant region and product regions^{37,38}. We define these explicitly, in terms of the eigenstates $\hat{R}|R\rangle = R|R\rangle$. The double well PESs are centered around $R = 0$ with a reactant region in the left well $\hat{P}_{\text{left}} = \int_{-\infty}^0 dR|R\rangle\langle R|$ and a product region in the right well $\hat{P}_{\text{right}} = \mathbb{1} - \hat{P}_{\text{left}}$. The forward reaction rate $k_{\rightarrow}$ is obtained under the ansatz of the rate equation:

$$\dot{P}_{\text{left}}(t) = -P_{\text{left}}(t)k_{\rightarrow} + (1 - P_{\text{left}}(t))k_{\leftarrow} \quad (13)$$

by obtaining the time-series $P_{\text{left}}(t) = \text{Tr}[\hat{P}_{\text{left}}\rho(t)]$ and solving for $k_{\rightarrow}$. The backwards reaction rate is denoted by $k_{\leftarrow}$ and its proportionality to $k_{\rightarrow}$ is linked to the equilibrium distribution of the reaction coordinate $\frac{k_{\rightarrow}}{k_{\leftarrow}} = \frac{P_{\text{right}}^{\text{eq}}}{P_{\text{left}}^{\text{eq}}}$, with e.g. $P_{\text{left}}^{\text{eq}} = \lim_{t \rightarrow \infty} P_{\text{left}}(t) = \frac{\text{Tr}[\hat{P}_{\text{left}}e^{-\beta H}]}{\text{Tr}[e^{-\beta H}]}$ ³⁹.

In the case of PESs with more than two wells, we define a projection operator for each well and solve for the reaction rate between neighboring wells. Further details are found in SI. sec. E.

In order to study the dynamics in more detail, we utilize the approach of^{40,41} to decompose the dynamics of the population of the vibrational states in terms of probability currents from a state n to state m :

$$\dot{\rho}_{nn}(t) = - \sum_m j_{n \rightarrow m}(t) \quad (14)$$

We further decompose each current $j_{n \rightarrow m}$ in terms of the contributions from the system Hamiltonian and contributions from the harmonic environment. We will mainly use this diagnostic tool in the case of the simple spectral density Eq. 12, such that we make the distinction between the following three contributions:

$$j_{n \rightarrow m}(t) = j_{n \rightarrow m}^{\text{Tunneling}}(t) + j_{n \rightarrow m}^{\text{Solvent}}(t) + j_{n \rightarrow m}^{\text{Cavity}}(t) \quad (15)$$

This allows us to directly interpret how the cavity affects the quantum dynamics and furthermore the role of the tunneling dynamics and the phonon bath. We also note that the reaction rate can be related to the probability currents that go from the reactant region to the product region (SI sec. F 1)

$$k_{\rightarrow} \approx \sum_n^{\text{reactant}} \sum_m^{\text{product}} j_{n \rightarrow m}(t) \quad (16)$$

This applies when the chosen basis is such, that the sum n runs over states localized in the left well and m indexes states localized the right well and when considering times t for which e.g. $P_{\text{left}}(t)$ has not changed considerably.

In the perturbative limit, the bath contributions to Eq. 14 can be reduced to the Pauli-equation

$$j_{n \rightarrow m}(t) \sim \Gamma_{n \rightarrow m} \rho_{nn}(t) - \Gamma_{m \rightarrow n} \rho_{mm}(t) \quad (17)$$

with the transition rates $\Gamma_{n \rightarrow m}$ given by Fermi's Golden Rule (SI. sec. F 2). These two reductions (Eq. 16 & 17) provide a simple perspective on how to relate the cavity modified noise spectrum to the cavity modified reaction rates, in a perturbative setting. Such a perspective has already been successfully applied in references^{20,25-27,42,43}. For the definition of the currents (Eq. 15) in terms of the master equation presented in the next section (Eq. 18), see the SI. sec. F.

III. METHODOLOGY

The time-series for Eq. 13 is obtained from a numerically exact simulation of the open quantum system dynamics using the methodology known under the names of HEOM⁴⁴, pseudomode^{45,46} and QD-MESS⁴⁷⁻⁴⁹, among others⁵⁰⁻⁵³. The main principle behind these methods is to use Markovian embeddings to emulate the effect of a

macroscopic bath through a reduced, few-body harmonic environment.

For the simulations in this study, the equations of motions for the few-body system $\rho(t)$ is:

$$\begin{aligned} \dot{\rho}(t) = & -i \left[\hat{H}_R + \lambda \hat{R}^2, \rho(t) \right] + L \rho(t) L^\dagger - \frac{1}{2} \{ L^\dagger L, \rho(t) \} \\ & + \sum_{k \in \mathcal{K}} \left\{ -z_k \hat{a}_k^\dagger \hat{a}_k \rho(t) - i \sqrt{d_k^*} \hat{R} \rho(t) \hat{a}_k^\dagger \right. \\ & \left. - i \sqrt{d_k} \left(\hat{a}_k + \hat{a}_k^\dagger \right) \hat{R} \rho(t) + H.C. \right\} \quad (18) \end{aligned}$$

where $H.C.$ refers to the hermitian conjugate of everything inside the brackets. $\hat{a}_k$ is the lowering operator of the k 'th auxiliary mode and z_k and d_k are complex parameters that characterizes the effective few-body harmonic environment. We have introduced the system jump operator L , in order to more efficiently emulate the environment, as will be elaborated upon in the following. The so-called reorganization energy λ is given by:

$$\lambda = -\text{Im} \sum_{k \in \mathcal{K}} \frac{d_k}{z_k} \quad (19)$$

and the initial state and system density matrix is given by:

$$\rho(0) = \rho_S(0) \otimes \overbrace{|0\rangle\langle 0| \otimes \dots \otimes |0\rangle\langle 0|}^{|\mathcal{K}| \text{ times}} =: \rho_S(0) \otimes^{\mathcal{K}} |0\rangle\langle 0| \quad (20)$$

$$\rho_S(t) = \text{Tr}_{\mathcal{K}} \left[\mathbb{1} \otimes^{\mathcal{K}} |0\rangle\langle 0| \rho(t) \right] \quad (21)$$

That is, the auxiliary modes are initialized in the vacuum and to obtain the reduced density matrix of the system, the auxiliary modes should be projected onto their vacuum. While $\mathcal{K}$ is the set that labels the effective bath modes of Eq. 18, consider an effective harmonic environment of K harmonic oscillators, similar to those in Eq. 18. These give rise to a bath correlation function (BCF) that is a sum of complex exponentials⁴⁷. If this effective BCF resembles the original BCF:

$$C(t) \approx \sum_k^K d_k e^{-z_k t} \quad d_k, z_k \in \mathbb{C} \quad (22)$$

the few-body system successfully emulates the macroscopic environment with deviations in observables being controlled by the decomposition in Eq. 22⁵⁴⁻⁵⁶, barring errors from numerically solving the master equation. Thus, the starting point of our method is to decompose the original BCF to machine precision, as a sum of complex exponentials. This is achieved using the AAA algorithm^{57,58}. Details and general considerations regarding the method or fitting procedure can be found in the SI. sec. D. Once the K pairs of z_k and d_k have been obtained, we reduce K to the smaller set $\mathcal{K}$ in the following manner.

1. Controlled compression of Markovian bath modes

In the literature^{44–49}, a HEOM-type master equation, such as Eq. 18, generally does not include system jump operators L . The inclusion of L in Eq. 18 stems from our novel approach to compress the number of effective bath modes needed to accurately simulate the open system dynamics, in a rigorous fashion. Essentially, some of the K harmonic oscillators resemble a Markovian bath. Thus, simulating them as an explicit degree of freedom is not necessary. This is made precise with the use of the Universal Lindblad Equation⁵⁹. As we show in the SI. sec. D 2, the auxiliary modes for which:

$$8 \sum_{k \in M} \frac{|d_k|}{\text{Re}(z_k)^2} \ll 1 \quad (23)$$

can instead be accounted for via

$$L = \sum_{n,m} \sqrt{S_M(E_m - E_n)} R_{nm} |n\rangle \langle m| \quad (24)$$

$$S_M(\omega) = \sum_{k \in M} 2 \text{Re} \left(\frac{d_k}{z_k - i\omega} \right) \quad (25)$$

where M is the set of harmonic oscillators which, according to Eq. 23, are deemed to be Markovian and $\hat{H}_R + \lambda \hat{R}^2 = \sum_n E_n |n\rangle \langle n|$. The set $\mathcal{K} = K \setminus M$ contains the remaining elements of K when the set M has been removed. This allows us to fit the BCF to machine precision (Eq. 22), while compressing the Hilbert space in Eq. 18 for the ensuing numerical simulations, which are carried out using the tensor network algorithm TDVP⁶⁰ implemented in ITensor^{61,62}.

IV. RESULTS AND DISCUSSIONS

The model for the physical system is primarily characterized by the two scalar functions $V(\hat{R})$, for the PES, and $J(\omega)$, for the effective spectral density. Therefore, the results section is divided into two parts, which aim to elucidate the role of each in the resulting reaction kinetics.

A. PES-driven effects

In the following, the environment will be assumed to consist of a phonon bath and a lossy cavity mode (Eq. 12). We utilize our parameterization of the PES to control and add additional transition frequencies to the reaction pathway. The probability currents between vibrational states, decomposed and assigned to their different contributors (Eq. 15), will then be used to analyze the dynamics.

1. Symmetric double well potential

We will begin by considering the simple symmetric double well, whose PES and vibrational states are shown in Fig. 4a. Outside the cavity, the reaction coordinate only interacts with the phonon bath. The strength of the coupling between the reaction coordinate and the solvent puts the system in the energy-diffusion limited regime (SI. sec. J). Tunneling between the wells has been minimized by the solvent and the thermal fluctuations it provides to the reaction is minimal, i.e. the reaction rate would increase by either increasing or decreasing the coupling to the phonon bath.

Still, as seen in Fig. 4c, the reaction outside the cavity is mainly driven by tunneling. Notice that the reaction rate, denoted by the horizontal line in Fig. 4b, is approximately given by the sum of rates for transitions going across the barrier (Eq. 16), which in this case is T_0 and T_1 . The transitions labeled a, b, c and d are included to provide the complete picture, despite their small values. Aside from diminishing tunneling, the thermal fluctuations of the phonon bath help the reaction coordinate transition over the potential energy barrier, which is captured by the transitions denoted by $T_\uparrow$ and $T_\downarrow$.

The cavity modified reaction rate is shown in Fig. 4b and the related probability currents are shown in Fig. 4c at different values of the cavity frequency, spanning resonant and off-resonant tunings. When resonant with the transition frequency of the wells $\omega_c = 1200 \text{cm}^{-1}$, the cavity is seen to greatly enhance both the excitation $T_\uparrow$ and relaxation $T_\downarrow$ processes. Furthermore, the tunneling through the top of the potential barrier T_0 is accordingly increased while the tunneling among the ground states T_0 is largely unaffected. Finally, when the cavity is tuned off resonance, its effect on the dynamics, as diagnosed by the probability currents, vanishes.

To elucidate these observations, we consider the time-scales of the system. Such an approach will also be helpful when we interpret the results of the coming sections. First, consider the timescale of the transition labeled T_1 . It must be much faster than T_0 since it constitutes tunneling the barrier at a point where it is narrower. Considering the transition T_0 itself provides a sizable portion of the reaction dynamics, despite its longer timescale, the transition T_1 must be far from being a rate-limiting step. Conversely, the rate at which spectral weight reaches this excited state manifold is much slower. To leading order, we can estimate the rate by Fermi's Golden Rule (Eq. 17) such that intra-well decay, due to the phonon bath, occurs at a rate $\sim \frac{8\lambda_R \gamma_R}{M\omega^2}$, while the excitation rate is related by detailed balance $\dots \times e^{-\beta\omega}$ where ω is the frequency of the well⁶³. Thus, excitation is the rate-limiting step, whose rate for the given values is of the order $\sim 10^{-6} \text{fs}^{-1}$ ⁶⁴. As a result, once the cavity is resonant with the molecular transition, the rate is strongly enhanced due to the increased value of the noise spectrum at the transition frequency²⁰. Effectively, the lossy cavity provides an additional thermalization channel, in

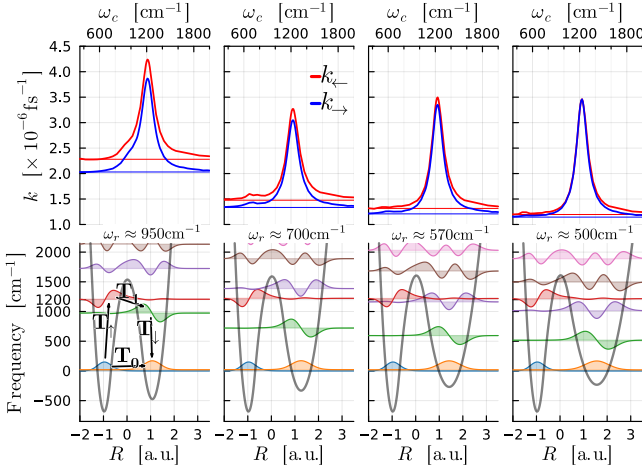


FIG. 5. The bottom row shows four different potential energy surfaces, with transition frequencies in the right-well of around $\omega_r \in \{950, 700, 570, 500\} \times \text{cm}^{-1}$. The top row shows the reaction rate from the left well to the right well in blue and the converse in red, for varying ω_c , when coupled to the spectral density in Eq. 12. The horizontal lines correspond to the reaction rates outside the cavity. The probability currents of the system with $\omega_r \approx 950 \text{cm}^{-1}$ are shown in Fig. 6.

provides an additional thermalization channel, will only have an effect on the reaction rate if it changes the rate-limiting step of the overall process. Excitations to the top of the potential energy barrier are still the rate-limiting step, so once the cavity is in resonance with one well, it means that the phonon-induced excitations of the other well becomes the rate-limiting step. Resultingly, the ratio between the magnitude of the primary peak and the secondary peak, in the reaction rate profiles, are related by the timescales provided in sec. IV A 1, such that for both the forward and backward reaction rates: $\frac{k(\omega_c=\omega_r)}{k(\omega_c=\omega_l)} \approx \frac{k_{\leftarrow}}{k_{\rightarrow}} \left(\frac{\omega_r}{\omega_l} \right)^2 e^{-\beta(\omega_l-\omega_r)}$ ⁶⁷. This explains the observed vanishing magnitude of the peak in k at ω_r when decreasing ω_r .

We also observe that the difference between the largest cavity-modified reaction rate and the reaction rate outside the cavity increases when widening the right well. This is attributable to the increased exposure to the phonon bath at lower frequencies. Once the cavity help facilitate the large frequency transition, the phonon bath can more easily assist the transitions of the lower frequency well, leading to a larger relative enhancement of the reaction rate for decreasing ω_r .

Figure 6 shows the probability currents of the transitions labeled in figure 5. Note that the probability currents are, by definition, anti-symmetric. Thus, if for a single dynamical simulation, we compared $T_{\downarrow}$ to its reversed partner, we would find that they only differ by a sign. However, here we compare two distinct dynamical simulations, that differ by which well the reaction coordinate wavefunction was initialized in. All the currents for the different systems in Fig. 5 can be found in the SI

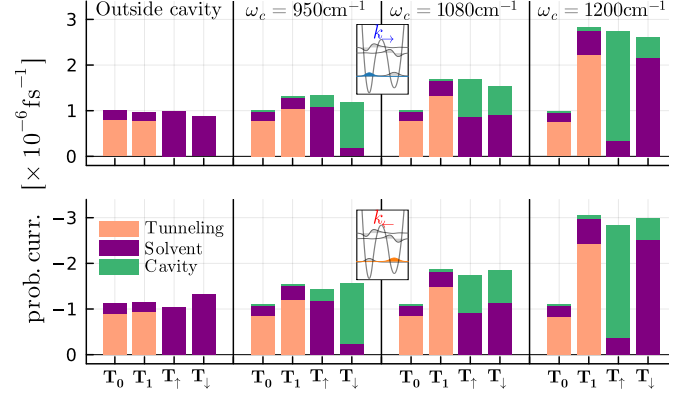


FIG. 6. The associated transition rates for both the $k_{\rightarrow}$ (top) and $k_{\leftarrow}$ (bottom) reaction dynamics of the least asymmetric system in figure 5. Transition rates for the other systems can be found in the SI Fig. 17.

Fig. 17.

Compared to the currents for the symmetric double well (Fig. 4c), the ground state tunneling rate T_0 is diminished, as already noted. From the currents, it is also apparent that the transition at 1200cm^{-1} , denoted $T_{\uparrow}$, is the rate-limiting step. Again, the tunneling along T_1 occurs at a faster time scale such that the net contribution simply follows what the thermodynamics of the wells dictates. At $\omega_c = \omega_r$ ($\approx 950 \text{cm}^{-1}$ for this PES) the cavity simply assumes the role of the phonon bath in enabling both excitation and relaxation transitions in the right well, as captured by $T_{\downarrow}$. How large a contribution either environmental dof provide to these transitions, is determined by the relative magnitude of their noise spectra (Fig. 3), in this perturbative regime. The more the cavity noise spectrum overlaps with the $\omega_l = 1200 \text{cm}^{-1}$ transition, the more the reaction dynamics are limited by the transitions in the right well, which occur on a faster time scale due to the $\sim \frac{1}{\omega^2}$ scaling of the phonon bath noise spectrum. This is apparent from the $T_{\downarrow}$ transitions in Fig. 6, whose solvent-induced component grows significantly when $\omega_c = 1200 \text{cm}^{-1}$.

3. Competing Pathways

Finally, we add a third relevant transition frequency to the dynamics. This will serve as our simplified model of a reaction coordinate containing two different product regions and thus two competing pathways.

Figure 7a shows the PES and the associated vibrational states. We define the middle well to be the reactant region and the left and right well to be two distinct product regions. The three near-degenerate states at frequency 1570cm^{-1} are not eigenstates of H_R , but instead have been mixed to construct the localized states seen (SI. sec. C3).

Outside the cavity, the reaction rate from the reactant to the right product region $k_{\rightarrow}$ is about $3.36 \times$ larger than

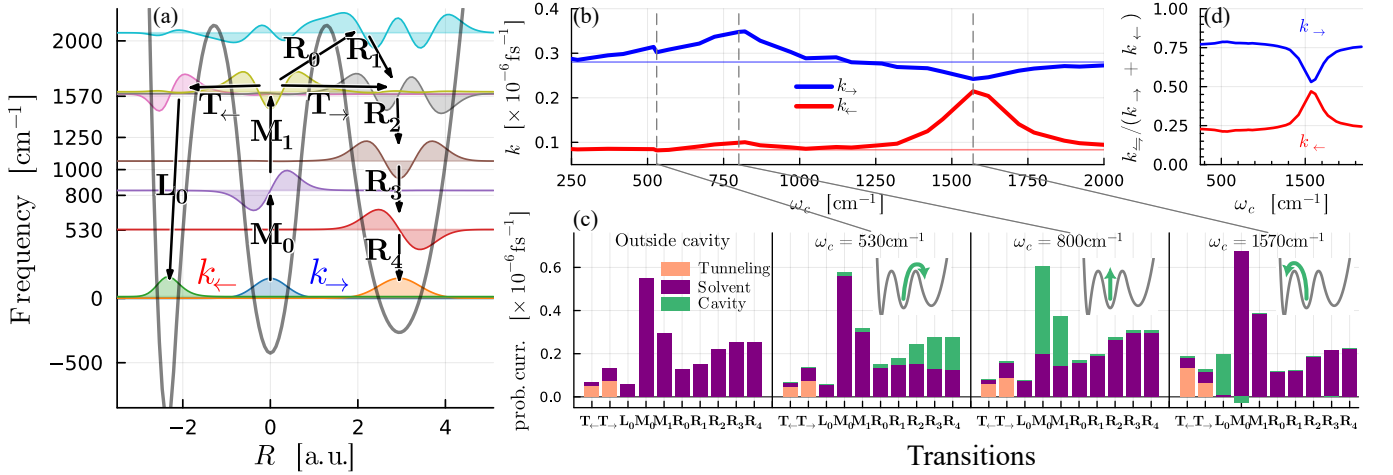


FIG. 7. (a) Visualization of the triple well PES and the vibrational states. The middle well is the reactant region and $k_{\leftarrow}$ and $k_{\rightarrow}$ denote the reaction rate to the left and right product wells, respectively. The black arrows label the state-to-state transitions. (b) Shows the reaction rates at different cavity frequencies. The horizontal lines denote the rates outside the cavity. (c) Probability currents for the associated transitions, both outside and inside the cavity. (d) The kinetic yield of the left product in red and the right product in blue for different cavity frequencies.

the reaction rate to the left product well $k_{\leftarrow}$, as seen in Fig. 7b. Informed by the probability currents in Fig. 7c, this difference in time scales can mainly be attributed to two factors. The right well ($\omega_r \approx 530 \text{ cm}^{-1}$) samples the phonon bath at a favorable frequency, leading to the increase in $T_{\rightarrow}$ over $T_{\leftarrow}$. Additionally, the right product well has an excited state, located in frequency just above 2000 cm^{-1} , that shares support with the reactant excited state leading to the phonon-assisted transition labeled R_0 , adding to the $k_{\rightarrow}$ reaction rate. Notice that $k_{\leftarrow}$ is approximately given by the probability current of the $T_{\leftarrow}$ transition and $k_{\rightarrow}$ by the $T_{\rightarrow}$ and R_0 transitions, in accordance with Eq. 16.

Inside the cavity, there exists three distinct scenarios depending on which of the three wells the cavity is resonant with. Similar to the examples with double-well PESs, we generally observe that the cavity accelerates the dynamics in the resonant well and that, due to the shape of the phonon bath, the higher frequency well generally hosts the rate-limiting steps.

When the cavity is resonant with the right product well $\omega_c = \omega_r$ the reaction rate $k_{\rightarrow}$ is slightly increased ($\sim 1.12\times$) whereas $k_{\leftarrow}$ is unaffected. The probability currents reflect this observation, where the cavity is seen to help thermalize the R_{0-4} transitions. Instead, when the cavity is resonant with the frequency of the reactant region $\omega_c = \omega_0 \approx 800 \text{ cm}^{-1}$ both reaction rates naturally increase, leading to an enhancement in $k_{\leftarrow}$ by $\sim 1.2\times$ and in $k_{\rightarrow}$ by $\sim 1.25\times$ in line with the cavity enhanced dynamics of the $M_{0,1}$ transitions.

More interestingly, when the cavity is resonant with the left product well $\omega_c = \omega_l$, the rate $k_{\leftarrow}$ is increased by $\sim 2.6\times$ whereas $k_{\rightarrow}$ is decreased by $\sim 0.86\times$, making the two reaction rates almost equal. The increased rate of equilibration between the left well and the reactant

well is expected, since the left-well frequency is resonant with the lossy cavity. The resulting cavity-induced bias toward the left well, reflected in the $T_{\leftarrow}$ and L_0 transitions, is accompanied by an increased excitation rate within the reactant well, as seen in the $M_{0,1}$ transitions. However, these effects do not fully compensate, so the excited reactant state feeding both product pathways is more depleted than without the cavity. This leads to the decrease in the R_{0-4} transitions and ultimately the $k_{\rightarrow}$ reaction rate.

The fact that the cavity can modify the two reaction rates differently can lead to cavity modified yield of the reaction. However, whether this occurs depends on the reversibility of the reaction. The system we have considered is best described as a reversible parallel reaction. The population in the middle well can flow into both the left and right well and the population in these outer wells can also flow back into the middle well. This reversibility allows the system to equilibrate and as such, the yield of the reaction is determined by the equilibrium distribution. Using the mean force Gibbs state^{39,66} already employed for the double well PESs (sec. IV A 2) we can see exactly how the cavity affects the equilibrium distribution. Similar to the double well examples, the solvent itself induces a small bias towards the more narrow and deeper left well but the cavity provides no noticeable effect (SI. sec. H 1). Thus, for the setup in question, the presence of the cavity does not change the yield of left and right product but it affects the speeds at which the reactant equilibrates with the two products.

A change in reaction rates can still induce a change in the yield, if the reaction is more accurately described as an irreversible parallel reaction. This is the case, for example, if the left and right product wells are much lower in energy than the reactant well, when compared to the

temperature. Then the backwards flow from the outer wells to the middle well would be negligible and the yield would be given by the branching ratios of the two forward reaction rates $P_{\text{left}}^{eq} = \frac{k_{\leftarrow}}{k_{\rightarrow} + k_{\leftarrow}}$ and $P_{\text{right}}^{eq} = \frac{k_{\rightarrow}}{k_{\rightarrow} + k_{\leftarrow}}$ (see the SI. sec. I for a mathematical formulation of this discussion). Thomas et al.⁵ provided the first experimental observation of such cavity-modified yield under kinetic control. Our model reproduces this effect, as seen in Fig. 7d, where the branching ratio is changed from being around 1 : 3 to 1 : 1 when the cavity is resonant with the frequency of the left well⁶⁸. Thus, the cavity can change product yields under kinetic control, by modifying the relative rates of competing pathways, but it does not significantly alter the equilibrium product distribution for the thermodynamically controlled reversible reaction considered here.

B. Environment-driven effects

All calculations in the preceding section employed the simple spectral density given in Eq. 12. We now turn our attention to the general spectral density, Eq. 7, which describes a reaction coordinate interacting with a spectator mode, both of which are coupled to the cavity mode. Simultaneously, we simplify the system by considering the symmetric double well seen in Fig. 4a.

How strongly the cavity couples to either the reaction coordinate η_R or the spectator mode η_Q depends on the alignment between the dipole moments and the cavity polarization. We parameterize this in terms of two angles:

$$\eta_Q \propto \cos(\phi) \quad \text{and} \quad \eta_R \propto \cos(\theta - \phi) \quad (26)$$

where ϕ is the angle between the cavity polarization and the spectator mode's dipole moment and θ is the angle between the reaction-coordinate and spectator-mode dipoles. This parameterization allows us to isolate how the relative geometry of the two vibrational dipoles controls the effect of the cavity on the reaction rate.

To represent an ensemble of molecular orientations, we first calculate the average spectral density $\langle J(\omega) \rangle_\phi := \frac{1}{\pi} \int_0^\pi J(\omega) d\phi$ ⁶⁹ before performing the dynamical simulation. If rotational diffusion occurs on a timescale much shorter than the reaction, this is the appropriate order of operations. Additionally, in the present parameter regime, the rate extracted from dynamics using $\langle J(\omega) \rangle_\phi$ is close to the effective rate obtained from a single-exponential fit to the ensemble reactant decay, $\langle e^{-k(\phi)t} \rangle_\phi$, representing the rate obtained by observing an ensemble of molecules. See the SI. sec. G for a discussion. The resulting noise spectra are visualized in figure 8, with explicit expressions found in the SI. sec. B 2.

We consider a spectator mode with frequency $\omega_Q = 1170\text{cm}^{-1}$, slightly below the relevant system transition at around 1200cm^{-1} . Outside the cavity, the damped spectator mode produces a Lorentzian feature in the

noise spectrum. Introducing the cavity hybridizes the cavity and spectator mode, giving rise to features associated with two Rabi-split damped resonances distanced $\Omega_R \sim 200\text{cm}^{-1}$ apart. Since the reaction coordinate also couples directly to the cavity, the resulting spectrum develops asymmetric and non-Lorentzian features that may be understood as reflecting interference between the two hybridized decay channels. This structure depends on θ , which controls the relative signs and magnitudes of the corresponding dipole couplings.

Using our method described in sec. III, we decompose, to great accuracy, the noise spectra in Fig. 8, for different cavity frequencies, and perform simulations of the dynamics in order to extract the reaction rate $k := k_{\rightarrow} = k_{\leftarrow}$. Figure 9 shows k as a function of ω_c for different angles θ ⁷⁰. The black horizontal line denotes the reaction rate outside the cavity, while the circles show the cavity-modified rates. The solid lines represent the perturbative predictions obtained from the probability currents. We now derive the corresponding closed expression for k , drawing on the analysis of Fig. 4.

According to Eq. 16, the reaction rate can be approximated by the total probability current across the potential barrier, $k \approx j_{T_0} + j_{T_1}$. The currents shown in Fig. 4c suggest that j_{T_0} is largely unaffected by the cavity, whereas the cavity-induced change in j_{T_1} originates from its contribution to the $T_{\uparrow,\downarrow}$ transitions. We may therefore approximate the cavity-modified reaction rate as the rate outside the cavity, k_0 , plus the cavity-induced contribution to these transitions. In the perturbative limit, Eq. 17 allows this contribution to be evaluated using Fermi's Golden Rule, yielding⁷¹:

$$k \approx k_0 + \text{constant} \times \int_{-\infty}^{\infty} d\omega e^{-\beta\omega} S_\beta^{(c)}(\omega) \mathcal{A}(\omega) d\omega \quad (27)$$

$S_\beta^{(c)}(\omega)$ refers to the cavity-modified component of the noise spectrum and $\mathcal{A}(\omega) = \frac{1}{\pi} \frac{\Gamma}{(\omega - \omega_0)^2 + \Gamma^2}$ encodes the broadening of the transition frequency due to the phonon bath²⁰ (see also SI. sec. F 2 a). Such expressions for the reaction rate were successfully employed in references^{20,72}, for the case of no spectator mode ($C_Q = \eta_Q = 0$) and in ref.²⁷, for no direct coupling between the reaction coordinate and the cavity ($\eta_R = 0$). Here, we have formulated a rationale in terms of probability currents and applied it to the case where all couplings are non-zero.

As seen in Fig. 9, the FGR predictions agree well with the HEOM simulations. Such agreement is not expected in general, since the assumptions underlying FGR require sufficiently weak bath coupling and a sufficiently smooth noise spectrum for the Born–Markov approximation to be applicable. Furthermore, the agreement in Fig. 9 requires an empirical rescaling of the cavity-induced FGR contribution. For Eq. 27 to be applicable in the first place, the system must also contain a well-defined rate-limiting step that can be identified.

Nonetheless, the qualitative form of the rate profiles in

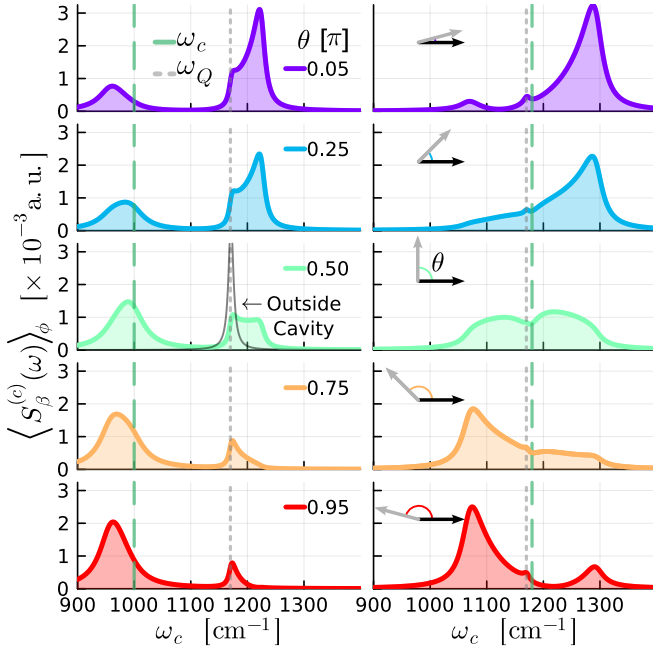


FIG. 8. Visualization of the noise spectra with spectral density given by Eq. 7, without the phonon bath, which has been averaged over the orientation of the molecule. The spectator mode frequency is $\omega_Q = 1170\text{cm}^{-1}$ with a lifetime of about $\sim 500\text{fs}$. θ is the angle between the reaction coordinate and spectator mode dipole moments. Five different values of θ is shown vertically, while two different values of ω_c are shown horizontally.

Fig. 9 is governed by which parts of the structured noise spectrum overlap with the broadened transition function $\mathcal{A}(\omega)$. If the spectator mode is far detuned from the relevant transition, $|\omega_Q - \omega_0| \gg \Gamma$, the two hybridized channels do not contribute comparably within the transition window, and the asymmetric interference feature becomes less pronounced. The rate profile then approaches a simpler resonant cavity response, although its amplitude and specific shape can still depend on θ when the cavity is tuned sufficiently close to the spectator mode to hybridize appreciably, $|\omega_c - \omega_Q| \sim \Omega_R \sim 200\text{cm}^{-1}$. This residual angular dependence reflects the way the dipole geometry redistributes spectral weight between the hybridized resonances.

When the spectator mode lies within the broadened transition window, $|\omega_Q - \omega_0| \lesssim \Gamma$, both cavity-spectator hybrid modes can contribute to the same rate-limiting transition, making the overlap with $\mathcal{A}(\omega)$ sensitive to interference between the two channels.

This interpretation is made more concrete by the fact that the rate curves in Fig. 9 permit a unified phenomenological description in terms of a Fano-plus-Lorentzian profile⁷³. The Fano component captures the interference between competing hybridized channels within the broadened transition window²⁹, while the Lorentzian component represents the remaining resonant

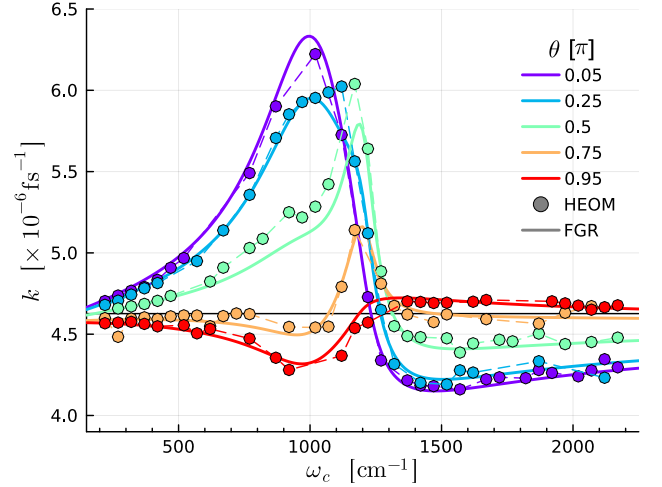


FIG. 9. Numerically exact reaction rates compared to predictions calculated via Eq. 27. The Lorentzian has parameters $\omega_0 = 1205\text{cm}^{-1}$ and $\Gamma = 30\text{cm}^{-1}$ and the constant includes an empirical rescaling factor of $\times 0.38$.

contribution from the spectator mode's residual spectral weight. In this form, the angle θ tunes the interference term through the Fano-parameter in addition to the magnitude of the residual Lorentzian contribution.

The rate profiles in Fig. 9 are therefore organized by an effective Fano-plus-Lorentzian structure, reflecting the interplay between cavity-spectator hybridization, spectral overlap with the broadened transition, and geometry-dependent interference.

V. CONCLUSION

We have developed a general quantum-dynamical framework for understanding cavity-modified reaction kinetics in terms of the individual transitions that comprise a reaction pathway. The framework combines a concise expression for the effective spectral density of an arbitrary harmonic environment with a controlled compression of Markovian auxiliary modes into system-level jump operators, enabling efficient tensor-network simulations of HEOM dynamics while retaining a physically transparent description of the environment.

The probability-current decomposition provides the central diagnostic for connecting these dynamics to observable reaction rates. It shows that cavity resonance selectively modifies particular transitions and that the impact on the overall reaction depends on their role in the reaction network. In systems with multiple transition frequencies, this distinction explains why resonances can differ greatly in their effect on the total rate and, in competing pathways, can redistribute flux between products.

For the competing pathways, the cavity modifies the rates of population transfer and hence the timescale of equilibration between the wells, but does not modify the

equilibrium distribution itself in the parameter regime considered here. The latter is governed by the mean force Gibbs state, whereas the cavity-induced changes in the transition rates redistribute flux along the way to equilibrium. Under irreversible conditions, where the final product distribution is instead determined kinetically by the relative pathway rates, this redistribution can translate directly into a cavity-modified product yield.

Finally, when spectator modes are included, hybridization and interference between environmental decay channels produce asymmetric, structured rate profiles that can be described in terms of Fano-plus-Lorentzian features. Taken together, these results show that cavity-modified reactivity is governed by how the cavity reshapes transition-resolved flux within a reaction network, rather than by a uniform modification of the reaction as a whole.

An important direction for future work is to extend this picture beyond the perturbative regime explored here. In the present calculations, the probability currents are consistent with the transition-rate intuition provided by Fermi's Golden Rule, while being obtained directly from the full HEOM dynamics rather than relying on a perturbative rate description. This makes the current-based analysis particularly promising for regimes where such rate theories become inadequate, as it provides a direct way to resolve how environmental effects propagate through the underlying reaction pathways. Extending the approach to more complex potential-energy landscapes, stronger system–environment and light–matter couplings, and more realistic models of the electromagnetic environment will help establish which of the mechanisms identified here persist beyond the present model systems. Importantly, the combination of general harmonic–environment representations, Markovian-mode compression, and tensor-network HEOM dynamics developed here provides a practical foundation for carrying out this analysis in increasingly complex settings.

ACKNOWLEDGMENTS

JV gratefully acknowledges the support of the Flatiron Institute during the Pre-Doctoral Researcher Program. JV also acknowledge support from the Carlsberg Foundation, and the Novo Nordisk Foundation, Grant No. NNF22SA0081175, Quantum Computing Programme.

The data and the code for reproducing the figures can be accessed on GitHub⁷⁴

¹T. W. Ebbesen, A. Rubio, and G. D. Scholes, "Introduction: Polaritonic chemistry," *Chemical Reviews* **123**, 12037–12038 (2023).

²R. Bhuyan, J. Mony, O. Kotov, G. W. Castellanos, J. Gómez Rivas, T. O. Shegai, and K. Börjesson, "The rise and current status of polaritonic photochemistry and photophysics," *Chemical Reviews* **123**, 10877–10919 (2023), pMID: 37683254.

³B. Xiang and W. Xiong, "Molecular polaritons for chemistry, photonics and quantum technologies," *Chemical Reviews* **124**, 2512–2552 (2024).

⁴J. A. Hutchison, T. Schwartz, C. Genet, E. Devaux, and T. W. Ebbesen, "Modifying chemical landscapes by coupling to vacuum fields," *Angewandte Chemie International Edition* **51**, 1592–1596 (2012).

⁵A. Thomas, L. Lethuillier-Karl, K. Nagarajan, R. M. A. Vergauwe, J. George, T. Chervy, A. Shalabney, E. Devaux, C. Genet, J. Moran, and T. W. Ebbesen, "Tilting a ground-state reactivity landscape by vibrational strong coupling," *Science* **363**, 615–619 (2019).

⁶J. Lather, P. Bhatt, A. Thomas, T. W. Ebbesen, and J. George, "Cavity catalysis by cooperative vibrational strong coupling of reactant and solvent molecules," *Angewandte Chemie International Edition* **58**, 10635–10638 (2019).

⁷A. Thomas, A. Jayachandran, L. Lethuillier-Karl, R. M. Vergauwe, K. Nagarajan, E. Devaux, C. Genet, J. Moran, and T. W. Ebbesen, "Ground state chemistry under vibrational strong coupling: dependence of thermodynamic parameters on the rabi splitting energy," *Nanophotonics* **9**, 249–255 (2020).

⁸W. Ahn, J. F. Triana, F. Recabal, F. Herrera, and B. S. Simpkins, "Modification of ground-state chemical reactivity via light–matter coherence in infrared cavities," *Science* **380**, 1165–1168 (2023).

⁹J. Lather, A. N. K. Thabassum, J. Singh, and J. George, "Cavity catalysis: modifying linear free-energy relationship under cooperative vibrational strong coupling," *Chemical Science* **13**, 195–202 (2022).

¹⁰F. Verdelli, Y.-C. Wei, K. Joseph, M. S. Abdelkhalik, M. Goudarzi, S. H. C. Askes, A. Baldi, E. W. Meijer, and J. Gomez Rivas, "Polaritonic chemistry enabled by non-local metasurfaces," *Angewandte Chemie International Edition* **63**, e202409528 (2024).

¹¹A. Thomas, J. George, A. Shalabney, M. Dryzhakov, S. J. Varma, J. Moran, T. Chervy, X. Zhong, E. Devaux, C. Genet, J. A. Hutchison, and T. W. Ebbesen, "Ground-state chemical reactivity under vibrational coupling to the vacuum electromagnetic field," *Angewandte Chemie International Edition* **55**, 11462–11466 (2016).

¹²D. M. Coles, N. Somaschi, P. Michetti, C. Clark, P. G. Lagoudakis, P. G. Savvidis, and D. G. Lidzey, "Polariton-mediated energy transfer between organic dyes in a strongly coupled optical microcavity," *Nature Materials* **13**, 712–719 (2014).

¹³X. Zhong, T. Chervy, L. Zhang, A. Thomas, J. George, C. Genet, J. A. Hutchison, and T. W. Ebbesen, "Energy transfer between spatially separated entangled molecules," *Angewandte Chemie International Edition* **56**, 9034–9038 (2017).

¹⁴B. Munkhbat, M. Wersäll, D. G. Baranov, T. J. Antosiewicz, and T. Shegai, "Suppression of photo-oxidation of organic chromophores by strong coupling to plasmonic nanoantennas," *Science Advances* **4**, eaas9552 (2018).

¹⁵K. Hirai, R. Takeda, J. A. Hutchison, and H. Uji-i, "Modulation of prins cyclization by vibrational strong coupling," *Angewandte Chemie International Edition* **59**, 5332–5335 (2020).

¹⁶S. Hou, M. Khatoniar, K. Ding, Y. Qu, A. Napolov, V. M. Menon, and S. R. Forrest, "Ultralong-range energy transport in a disordered organic semiconductor at room temperature via coherent exciton-polariton propagation," *Advanced Materials* **32**, 2002127 (2020).

¹⁷A. Sau, K. Nagarajan, B. Patraha, L. Lethuillier-Karl, R. M. A. Vergauwe, A. Thomas, J. Moran, C. Genet, and T. W. Ebbesen, "Modifying woodward–hoffmann stereoselectivity under vibrational strong coupling," *Angewandte Chemie International Edition* **60**, 5712–5717 (2021).

¹⁸S. Satapathy, M. Khatoniar, D. K. Parappuram, B. Liu, G. John, J. Feist, F. J. Garcia-Vidal, and V. M. Menon, "Selective isomer emission via funneling of exciton polaritons," *Science Advances* **7**, eabj0997 (2021).

¹⁹J. Singh, P. Garg, J. Lather, and J. George, "Stirring chemical reaction landscapes through strong light–matter interactions," *Chemical Communications* **61**, 18289–18301 (2025).

²⁰W. Ying and P. Huo, "Resonance theory and quantum dynamics

- simulations of vibrational polariton chemistry,” *The Journal of Chemical Physics* **159**, 084104 (2023).
- ²¹W. Ying, M. E. Mondal, E. R. Koessler, S. M. Vega, and P. Huo, “Collective effects in polariton chemistry and photophysics,” *Annual Review of Physical Chemistry* **77**, 201–224 (2026).
 - ²²L. P. Lindoy, A. Mandal, and D. R. Reichman, “Quantum dynamical effects of vibrational strong coupling in chemical reactivity,” *Nature Communications* **14**, 2733 (2023).
 - ²³A. Mandal, M. A. Taylor, B. M. Weight, E. R. Koessler, X. Li, and P. Huo, “Theoretical advances in polariton chemistry and molecular cavity quantum electrodynamics,” *Chemical Reviews* **123**, 9786–9879 (2023).
 - ²⁴Y. Ke and J. O. Richardson, “Quantum nature of reactivity modification in vibrational polariton chemistry,” *The Journal of Chemical Physics* **161**, 054104 (2024).
 - ²⁵W. Ying, M. A. D. Taylor, and P. Huo, “Resonance theory of vibrational polariton chemistry at the normal incidence,” *Nanophotonics* **13**, 2601–2615 (2024).
 - ²⁶W. Ying and P. Huo, “Resonance theory of vibrational strong coupling enhanced polariton chemistry and the role of photonic mode lifetime,” *Communications Materials* **5**, 110 (2024).
 - ²⁷S. M. Vega, W. Ying, and P. Huo, “Theoretical insights into the resonant suppression effect in vibrational polariton chemistry,” *Journal of the American Chemical Society* **147**, 19727–19737 (2025).
 - ²⁸Y. Ke, “Stochastic resonance in vibrational polariton chemistry,” *The Journal of Chemical Physics* **162**, 064702 (2025).
 - ²⁹Y. Ke, “Cooperative chemical reactions in optical cavities: A complex interplay of mode hybridization, timescale balance, and pathway interference,” *The Journal of Chemical Physics* **164**, 174103 (2026).
 - ³⁰L. P. Lindoy, A. Mandal, and D. R. Reichman, “Investigating the collective nature of cavity-modified chemical kinetics under vibrational strong coupling,” *Nanophotonics* **13**, 2617–2633 (2024).
 - ³¹M. Ruggenthaler, D. Sidler, and A. Rubio, “Understanding polaritonic chemistry from ab initio quantum electrodynamics,” *Chemical Reviews* **123**, 11191–11229 (2023).
 - ³²A. Caldeira and A. Leggett, “Quantum tunnelling in a dissipative system,” *Annals of Physics* **149**, 374–456 (1983).
 - ³³R. Feynman and F. Vernon, “The theory of a general quantum system interacting with a linear dissipative system,” *Annals of Physics* **24**, 118–173 (1963).
 - ³⁴H.-P. Breuer and F. Petruccione, *The Theory of Open Quantum Systems* (Oxford University Press, 2007).
 - ³⁵A. Garg, J. N. Onuchic, and V. Ambegaokar, “Effect of friction on electron transfer in biomolecules,” *The Journal of Chemical Physics* **83**, 4491–4503 (1985).
 - ³⁶Following ref.²⁰, we write $\alpha(\omega_c) = \frac{1-e^{-\beta\omega_c}}{\tau_c}$, which is more or less constant $\alpha \approx \tau_c^{-1}$ for the $\beta\omega_c$ considered here.
 - ³⁷W. H. Miller, S. D. Schwartz, and J. W. Tromp, “Quantum mechanical rate constants for bimolecular reactions,” *The Journal of Chemical Physics* **79**, 4889–4898 (1983).
 - ³⁸I. R. Craig, M. Thoss, and H. Wang, “Proton transfer reactions in model condensed-phase environments: Accurate quantum dynamics using the multilayer multiconfiguration time-dependent hartree approach,” *The Journal of Chemical Physics* **127**, 144503 (2007).
 - ³⁹A. S. Trushechkin, M. Merkli, J. D. Cresser, and J. Anders, “Open quantum system dynamics and the mean force gibbs state,” *AVS Quantum Science* **4**, 012301 (2022).
 - ⁴⁰J. J. Roden and K. B. Whaley, “Probability-current analysis of energy transport in open quantum systems,” *Phys. Rev. E* **93**, 012128 (2016).
 - ⁴¹B. Schumacher, M. D. Westmoreland, A. New, and H. Qiao, “Probability current and thermodynamics of open quantum systems,” (2016).
 - ⁴²J. E. Lawrence, T. Fletcher, L. P. Lindoy, and D. E. Manolopoulos, “On the calculation of quantum mechanical electron transfer rates,” *The Journal of Chemical Physics* **151**, 114119 (2019).
 - ⁴³Y. Lai, W. Ying, and P. Huo, “Non-equilibrium rate theory for polariton relaxation dynamics,” *The Journal of Chemical Physics* **161**, 104109 (2024).
 - ⁴⁴Y. Tanimura, “Numerically “exact” approach to open quantum dynamics: The hierarchical equations of motion (heom),” *The Journal of Chemical Physics* **153**, 020901 (2020).
 - ⁴⁵G. Pleasance, B. M. Garraway, and F. Petruccione, “Generalized theory of pseudomodes for exact descriptions of non-markovian quantum processes,” *Phys. Rev. Res.* **2**, 043058 (2020).
 - ⁴⁶G. Park, Z. Huang, Y. Zhu, C. Yang, G. K.-L. Chan, and L. Lin, “Quasi-lindblad pseudomode theory for open quantum systems,” *Phys. Rev. B* **110**, 195148 (2024).
 - ⁴⁷M. Xu, Y. Yan, Q. Shi, J. Ankerhold, and J. T. Stockburger, “Taming quantum noise for efficient low temperature simulations of open quantum systems,” *Phys. Rev. Lett.* **129**, 230601 (2022).
 - ⁴⁸M. Xu, V. Vadimov, M. Krug, J. T. Stockburger, and J. Ankerhold, “A universal framework for quantum dissipation: minimally extended state space and exact time-local dynamics,” (2023).
 - ⁴⁹M. Xu, V. Vadimov, J. T. Stockburger, and J. Ankerhold, “Colloquium: Simulating non-markovian dynamics in open quantum systems,” *Rev. Mod. Phys.* **98**, 021002 (2026).
 - ⁵⁰K. H. Hughes, C. D. Christ, and I. Burghardt, “Effective-mode representation of non-markovian dynamics: A hierarchical approximation of the spectral density. i. application to single surface dynamics,” *The Journal of Chemical Physics* **131**, 024109 (2009).
 - ⁵¹A. W. Chin, Á. Rivas, S. F. Huelga, and M. B. Plenio, “Exact mapping between system-reservoir quantum models and semi-infinite discrete chains using orthogonal polynomials,” *Journal of Mathematical Physics* **51**, 092109 (2010).
 - ⁵²M. P. Woods, R. Groux, A. W. Chin, S. F. Huelga, and M. B. Plenio, “Mappings of open quantum systems onto chain representations and markovian embeddings,” *Journal of Mathematical Physics* **55**, 032101 (2014).
 - ⁵³A. Nazir and G. Schaller, “The reaction coordinate mapping in quantum thermodynamics,” in *Thermodynamics in the Quantum Regime: Fundamental Aspects and New Directions*, edited by F. Binder, L. A. Correa, C. Gogolin, J. Anders, and G. Adesso (Springer International Publishing, Cham, 2018) pp. 551–577.
 - ⁵⁴F. Mascherpa, A. Smirne, S. F. Huelga, and M. B. Plenio, “Open systems with error bounds: Spin-boson model with spectral density variations,” *Phys. Rev. Lett.* **118**, 100401 (2017).
 - ⁵⁵Z. Huang, Y. Zhu, G. Park, and L. Lin, “Unified error bounds for perturbations of non-markovian open quantum systems in gaussian environments,” (2025).
 - ⁵⁶K. Liu and J. Lu, “Error Bounds for Open Quantum Systems with Harmonic Bosonic Bath,” *Quantum* **9**, 1896 (2025).
 - ⁵⁷T. A. Driscoll, Y. Nakatsukasa, and L. N. Trefethen, “Aaa rational approximation on a continuum,” *SIAM Journal on Scientific Computing* **46**, A929–A952 (2024).
 - ⁵⁸T. A. Driscoll, “Rational function approximation in julia,” (2025).
 - ⁵⁹F. Nathan and M. S. Rudner, “Universal lindblad equation for open quantum systems,” *Phys. Rev. B* **102**, 115109 (2020).
 - ⁶⁰J. Haegeman, J. I. Cirac, T. J. Osborne, I. Pižorn, H. Verschelde, and F. Verstraete, “Time-dependent variational principle for quantum lattices,” *Phys. Rev. Lett.* **107**, 070601 (2011).
 - ⁶¹M. Fishman, S. R. White, and E. M. Stoudenmire, “The ITensor Software Library for Tensor Network Calculations,” *SciPost Phys. Codebases*, 4 (2022).
 - ⁶²M. Fishman, S. R. White, and E. M. Stoudenmire, “Codebase release 0.3 for ITensor,” *SciPost Phys. Codebases*, 4–r0.3 (2022).
 - ⁶³At the relevant transition frequencies $\omega > \gamma_R = 200\text{cm}^{-1}$ the phonon bath becomes $\frac{2\lambda_R\gamma_R\omega}{\omega^2+\gamma_R^2} \approx \frac{2\lambda_R\gamma_R}{\omega}$ and the transition matrix element within a well is approximated as $|\langle 1|\hat{R}|0\rangle|^2 \approx \frac{2}{M\omega}$ with $|n\rangle$ the harmonic oscillator states. The noise spectrum evaluated at the transition frequency times the matrix element, yields the FGR rate which are the quoted expressions. Alternatively, see SI. sec. F 2.
 - ⁶⁴ $\frac{\lambda\gamma}{\omega^2} e^{-\omega/T} \frac{8}{M} \approx \frac{83.6\text{cm}^{-1} \times 200\text{cm}^{-1}}{1200\text{cm}^{-1}} e^{-1200\text{cm}^{-1}/208.5\text{cm}^{-1}} \times$

- $0.09\text{fs}^{-1} \approx 3.31 \times 10^{-6}\text{fs}^{-1}$.
- ⁶⁵B. Metzger, E. Muller, J. Nishida, B. Pollard, M. Hentschel, and M. B. Raschke, "Purcell-enhanced spontaneous emission of molecular vibrations," *Phys. Rev. Lett.* **123**, 153001 (2019).
- ⁶⁶J. D. Cresser and J. Anders, "Weak and ultrastrong coupling limits of the quantum mean force gibbs state," *Phys. Rev. Lett.* **127**, 250601 (2021).
- ⁶⁷Using that $k_{\rightarrow} = \frac{P_{\text{right}}^{eq}}{P_{\text{left}}^{eq}} k_{\leftarrow}$, the rate $k_{\rightarrow}(\omega_c = \omega_l)$ can be bound by the phonon-induced excitations $k_{\rightarrow}(\omega_c = \omega_l) \sim \frac{P_{\text{right}}^{eq}}{P_{\text{left}}^{eq}} \frac{e^{-\beta\omega_r}}{\omega_r^2}$. Conversely, $k_{\rightarrow}(\omega_c = \omega_r)$ is bounded by the phonon-induced excitations in the left well, such that the ratio leads to the quoted result.
- ⁶⁸Whether the cavity-modification is enhancive or suppressive in nature, is largely dependent on the harmonic environment²². Thus, if we instead considered the coupling between the reaction coordinate and the cavity to be mediated by a spectator mode²⁷, we would expect the presence of the cavity to have a suppressive effect, which could place our model more in line with the experimental study⁵. Additionally, the potential energy of the product regions could be lowered to incorporate the irreversible aspect of the reaction.
- ⁶⁹This is the same kind of averaging used in²⁷ (see their SI). However, it does not correspond to isotropic averaging: $\frac{1}{2} \int_0^\pi \dots \sin(\phi) d\phi$, that some of the same authors applied in²⁰. Applying the isotropic average would have been more physically relevant, as it corresponds to a uniform distribution of the molecular orientation in three dimensional space. Assuming we can apply FGR to describe the cavity-modified reaction rates not only for the orientationally averaged spectral density, but also for each ϕ , we perform a comparison for the two different averages in the SI. sec. G. As anticipated, there is only a small quantitative difference between the two and the physically relevant qualitative behavior can still be inferred from the results presented in sec. IV B.
- ⁷⁰The HEOM results for some ω_c are omitted. This is because the automated AAA decomposition (Eq. 22) did not reach the specified accuracy of 10^{-10} . HEOM results for any ω_c and θ can still be obtained with our method, in some cases one would just have to tune the settings for the AAA-fit by hand or choose a different method to achieve the decomposition.
- ⁷¹The reverse transition in Eq. 17 is assumed negligible due to the small population in the excited state. Also, the constant refers to the absolute square of the matrix element between the localized ground and excited state, multiplied by an *ad hoc* fitting factor. In the case of Fig. 9 this scalar is 0.38.
- ⁷²W. Ying and P. Huo, "Resonance theory of vibrational strong coupling enhanced polariton chemistry and the role of photonic mode lifetime," *Communications Materials* **5**, 110 (2024).
- ⁷³Specifically, we have in mind:
- $$k(\tilde{\omega}_c) - k_0 \propto \frac{(a(\theta) + \tilde{\omega}_c)^2}{1 + \tilde{\omega}_c^2} + b(\theta) \frac{\tilde{\Gamma}}{(\tilde{\omega}_c + \tilde{\omega}_0)^2 + \tilde{\Gamma}^2}$$
- where $\tilde{\omega}_c$ is a dimensionless cavity frequency that has been shifted
- and scaled appropriately and $a(\theta)$ is the angle-dependent Fano-parameter. Reducing Eq. 27 to this form was not immediately successful because naive pole-expansion of the terms in Eq. B51 missed some of the observed features.
- ⁷⁴GitHub repository: <https://github.com/Vinther901/Vinther2026dynamics>.
- ⁷⁵J. Williamson, "The exponential representation of canonical matrices," *American Journal of Mathematics* **61**, 897–911 (1939).
- ⁷⁶F. Nicacio, "Williamson theorem in classical, quantum, and statistical physics," *American Journal of Physics* **89**, 1139–1151 (2021).
- ⁷⁷D. T. Colbert and W. H. Miller, "A novel discrete variable representation for quantum mechanical reactive scattering via the s-matrix kohn method," *The Journal of Chemical Physics* **96**, 1982–1991 (1992).
- ⁷⁸M. Krug and J. Stockburger, "On stability issues of the heom method," *The European Physical Journal Special Topics* **232**, 3219–3226 (2023).
- ⁷⁹M. Sánchez-Barquilla and J. Feist, "Accurate truncations of chain mapping models for open quantum systems," *Nanomaterials* **11** (2021), 10.3390/nano11082104.
- ⁸⁰L. K. Zhou, G. R. Jin, and W. Yang, "Systematic and efficient pseudomode method to simulate open quantum systems under a bosonic environment," *Phys. Rev. A* **110**, 022221 (2024).
- ⁸¹Z. Huang, G. Park, G. K.-L. Chan, and L. Lin, "Coupled lindblad pseudomode theory for simulating open quantum systems," *Phys. Rev. Lett.* **136**, 090403 (2026).
- ⁸²L. A. Correa and J. Glatthard, "Potential renormalisation, Lamb shift and mean-force Gibbs state – to shift or not to shift?" *Quantum* **9**, 1868 (2025).
- ⁸³J. Agerskov and F. Nathan, "Markovian quantum master equations are exponentially accurate in the weak coupling regime," (2026).
- ⁸⁴L. Chen and Q. Shi, "Quantum rate dynamics for proton transfer reactions in condensed phase: The exact hierarchical equations of motion approach," *The Journal of Chemical Physics* **130**, 134505 (2009).
- ⁸⁵If the trace-preserving version of the master equation (Eq. D1) was instead used, then $\rho_k(t) = \text{Tr}_B[\hat{a}_k \rho(t)]$.
- ⁸⁶A. Plonka, *Dispersive Kinetics* (Springer Netherlands, 2001).
- ⁸⁷E. Anslyn and D. Dougherty, *Modern Physical Organic Chemistry*, G - Reference, Information and Interdisciplinary Subjects Series (University Science Books, 2006).
- ⁸⁸C. Zhang, E. Jeckelmann, and S. R. White, "Density matrix approach to local hilbert space reduction," *Phys. Rev. Lett.* **80**, 2661–2664 (1998).
- ⁸⁹E. Jeckelmann and S. R. White, "Density-matrix renormalization-group study of the polaron problem in the holstein model," *Phys. Rev. B* **57**, 6376–6385 (1998).
- ⁹⁰M. S. Rudolph and J. Tindall, "Simulating and sampling from quantum circuits with 2d tensor networks," (2025), arXiv:2507.11424 [quant-ph].

SUPPLEMENTARY INFORMATION

A	Gaussian environments	14
B	Basis-independent spectral density for general harmonic environments	15
1	Spectral density of interacting lossy harmonic modes.	18
2	Spectral density of interacting cavity and spectator mode, averaged over molecule orientation	20

C	Reaction Coordinate Potentials & vibrational state details	20
1	Double Well	20
a	Derivation	21
2	Triple Well	21
3	Localized states	22
4	Numerics	22
D	Details on Hierarchical Equations Of Motion	23
1	Decomposing the Bath Correlation Function	24
2	Effective Markovian correction	25
E	Reaction Rate Analysis	26
1	Triple well reaction rates	27
F	Probability Currents: Transition Rate Between States	28
1	Relating probability-currents to reaction rates	29
2	FGR as perturbative limit of probability currents	30
a	Broadening of transition frequencies due to phonon bath	31
G	Distribution of reaction rates and comparison to isotropic averaging	31
H	Equilibrium state of the reaction coordinate	32
1	Mean force Gibbs state for the triple well, Fig. 7	33
2	Mean force Gibbs state for the double wells, Fig. 5	33
I	Kinetic and thermodynamic control of the triple well system	34
J	The phonon bath and Kramer's turnover	36
a	Tensor network simulations in the strong coupling regime	36
K	Additional asymmetric double well probability currents	37
L	Overview of parameter values	37

Supplemental Material A: Gaussian environments

A generic starting point for considering a system coupled to a harmonic environment, is the mass-weighted Caldeira-Legget model of dissipation³²:

$$\hat{H} = \hat{H}_S + \frac{1}{2} \sum_j \left(\hat{p}_j^2 + \omega_j^2 \left(\hat{x}_j - \frac{c_j}{\omega_j^2} \hat{S} \right)^2 \right) \quad (\text{A1})$$

The coupling $\frac{\omega_j^2}{2} \left(\hat{x}_j - \frac{c_j}{\omega_j^2} \hat{S} \right)^2 = \frac{\omega_j^2}{2} \hat{x}_j^2 - c_j \hat{x}_j \hat{S} + \frac{c_j^2}{2\omega_j^2} \hat{S}^2$ gives rise to the reorganization term: $\sum_j \frac{c_j^2}{2\omega_j^2} \hat{S}^2 =: \lambda \hat{S}^2$. It is important, in general, to include this term as it counters unwarranted bath-induced renormalization of the system. Consider e.g. two harmonic oscillators coupled in this manner. When the coupling is strong enough, the frequency of one of the hybrid modes will become negative, unless the reorganization term is not included. By assuming an initially factorized state and with the harmonic environment in a thermal equilibrium, one can formally write the system reduced density matrix³⁴:

$$\rho_S(t) = e^{\hat{\mathcal{L}}_S t} \mathcal{T}_S \exp \left[- \int_0^t dt_1 \int_0^{t_1} dt_2 \hat{\mathcal{K}}(t_1, t_2) \right] \rho_S(0) \quad (\text{A2})$$

with the influence superoperator

$$\hat{\mathcal{K}}(t_1, t_2) \bullet = \left(\hat{S}(t_1) \bullet - \bullet \hat{S}(t_1) \right) \left(C(t_1 - t_2) \hat{S}(t_2) \bullet - \bullet \hat{S}(t_2) C^*(t_1 - t_2) \right) \quad (\text{A3})$$

where $C(t_1 - t_2)$ is the bath correlation function (BCF). From the above, we see that the only information we need to model the environment is exactly the BCF. We may rewrite this in terms of the *spectral density* $J(\omega)$:

$$J(\omega) = \frac{\pi}{2} \sum_j \frac{c_j^2}{\omega_j} \delta(\omega - \omega_j) \quad (\text{A4})$$

such that:

$$\int_{-\infty}^{\infty} dt C(t) e^{i\omega t} = S_{\beta}(\omega) = \frac{2}{1 - e^{-\beta\omega}} (J(\omega) - J(-\omega)) \quad (\text{A5})$$

This expression can also be found in the literature, written with only one $J(\omega)$ with the understanding that $J(\omega)$ is already an odd function or else you employ the odd version $J(\omega) - J(-\omega)$.

Some remarks regarding the BCF and noise spectrum:

- $S_{\beta}(\omega)$ is real and positive and within Fermi's Golden Rule, the transition rate at frequency ω is proportional to $S_{\beta}(-\omega)$.
- $S_{\beta}(-\omega) = e^{-\beta\omega} S_{\beta}(\omega)$ known as the detailed balance condition.
- $C(t)$ is complex with:

$$C(t) = C^*(-t) \quad (\text{A6})$$

$$\text{Re}[C(t)] = \int_0^{\infty} \frac{d\omega}{\pi} \coth\left(\frac{\beta\omega}{2}\right) J(\omega) \cos(\omega t) \quad (\text{A7})$$

$$\text{Im}[C(t)] = - \int_0^{\infty} \frac{d\omega}{\pi} J(\omega) \sin(\omega t) \quad (\text{A8})$$

- The reorganization energy is given by $\sum_j \frac{c_j^2}{2\omega_j^2} = \int_0^{\infty} \frac{d\omega}{\pi} \frac{J(\omega)}{\omega} = -\text{Im} \int_0^{\infty} C(t) dt$.

Supplemental Material B: Basis-independent spectral density for general harmonic environments

The above model considered a collection of uncoupled harmonic oscillators, each coupled to the system. However, it is of interest to consider more general microscopic models where the interactions can be arbitrary. This will allow us to describe the entire harmonic environment, including cavity modes, spectator modes and so on, in terms of a single spectral density (or more if more system operators are coupled to the environment). For now, we will ignore the reorganization term and write the system-bath Hamiltonian as:

$$\mathcal{H} = H_S + \hat{S} \mathbf{c}^T \hat{\mathbf{r}} + \frac{1}{2} \hat{\mathbf{r}}^T H \hat{\mathbf{r}} \quad (\text{B1})$$

where $\hat{\mathbf{r}}$ is a vector containing the $\hat{x}_j$'s and $\hat{p}_j$'s and H is a positive-definite symmetric matrix that contains the parameters for the bath modes. $\mathbf{c}$ is the vector of coupling constants. We will calculate the BCF, which in this notation, is:

$$C(t) = \text{Tr}[\mathbf{c}^T \hat{\mathbf{r}} e^{-\frac{it}{2} \hat{\mathbf{r}}^T H \hat{\mathbf{r}}} \mathbf{c}^T \hat{\mathbf{r}} e^{\frac{it}{2} \hat{\mathbf{r}}^T H \hat{\mathbf{r}}} \rho_B(0)] \quad (\text{B2})$$

$\rho_B(0) \propto e^{-\frac{\beta}{2} \hat{\mathbf{r}}^T H \hat{\mathbf{r}}}$ is the initial thermal state of the bath. The goal is to check whether the above permits the same expression for the noise spectrum in terms of a spectral density as the Caldeira-Leggett model and to find a practical expression in terms of H and $\mathbf{c}$. The time-dependence can be simplified by considering:

$$\frac{1}{2} [\mathbf{c}^T \hat{\mathbf{r}}, \hat{\mathbf{r}}^T H \hat{\mathbf{r}}] = \frac{1}{2} c_n H_{ml} (\hat{r}_n \hat{r}_m \hat{r}_l - \hat{r}_m \hat{r}_l \hat{r}_n) \quad (\text{B3})$$

$$= \frac{1}{2} c_n H_{ml} ([\hat{r}_n, \hat{r}_m] \hat{r}_l - \hat{r}_m [\hat{r}_l, \hat{r}_n]) \quad (\text{B4})$$

$$= \frac{i}{2} \mathbf{c}^T \Omega H \hat{\mathbf{r}} - \frac{i}{2} \hat{\mathbf{r}}^T H \Omega \mathbf{c} = i \mathbf{c}^T \Omega H \hat{\mathbf{r}} \quad (\text{B5})$$

where we have defined the symplectic matrix $i\Omega_{nm} = [\hat{r}_n, \hat{r}_m]$. Then we can apply the Campbell identity:

$$e^{-\frac{it}{2} \hat{\mathbf{r}}^T H \hat{\mathbf{r}}} \mathbf{c}^T \hat{\mathbf{r}} e^{\frac{it}{2} \hat{\mathbf{r}}^T H \hat{\mathbf{r}}} = e^{\frac{it}{2} [\bullet, \hat{\mathbf{r}}^T H \hat{\mathbf{r}}]} \mathbf{c}^T \hat{\mathbf{r}} = \mathbf{c}^T e^{-t\Omega H \hat{\mathbf{r}}} \quad (\text{B6})$$

such that:

$$C(t) = \mathbf{c}^T \text{Tr}[\hat{\mathbf{r}} \hat{\mathbf{r}}^T \rho_B(0)] e^{tH\Omega \mathbf{c}} \quad (\text{B7})$$

where the trace is still in the Liouville-space sense and not in the matrix sense. Through the normal mode or Williamson decomposition^{75,76}, we may write $H = S^T D S$ where the diagonal matrix D contains the frequencies $\{\omega_j\}$ and S is a symplectic matrix ($S^T \Omega S = \Omega$). Then we can define the following Gaussian unitary:

$$\mathcal{U} \hat{\mathbf{r}} \mathcal{U}^\dagger = S \hat{\mathbf{r}} \quad \text{with} \quad \mathcal{U} = e^{-\frac{i}{2} \hat{\mathbf{r}}^T K \hat{\mathbf{r}}} \quad \text{and} \quad e^{\Omega K} = S \quad (\text{B8})$$

such that:

$$\text{Tr}[\hat{\mathbf{r}} \hat{\mathbf{r}}^T \rho_B(0)] = \text{Tr}[\mathcal{U}^\dagger \hat{\mathbf{r}} \hat{\mathbf{r}}^T \mathcal{U} \rho_B^D(0)] = S^{-1} \text{Tr}[\hat{\mathbf{r}} \hat{\mathbf{r}}^T \rho_B^D(0)] S^{-T} \quad (\text{B9})$$

where $\rho_B^D(0)$ is the thermal state of the diagonal modes:

$$\rho_B^D(0) \propto \prod_j e^{-\frac{\beta \omega_j}{2} (\hat{x}_j^2 + \hat{p}_j^2)} \quad (\text{B10})$$

Then, we can calculate the trace using that:

$$\text{Tr} \left[\begin{pmatrix} \hat{x}_j \\ \hat{p}_j \end{pmatrix} \begin{pmatrix} \hat{x}_j & \hat{p}_j \end{pmatrix} e^{-\frac{\beta \omega_j}{2} (\hat{x}_j^2 + \hat{p}_j^2)} \right] = \frac{1}{2} \begin{pmatrix} 2n(\omega_j) + 1 & i \\ -i & 2n(\omega_j) + 1 \end{pmatrix} \quad (\text{B11})$$

with $n(\omega_j) = 1/(e^{\beta \omega_j} - 1)$, such that we get:

$$\text{Tr}[\hat{\mathbf{r}} \hat{\mathbf{r}}^T \rho_B^D(0)] = \frac{1}{2} \left(i\Omega + \coth\left(\frac{\beta}{2} D\right) \right) \quad (\text{B12})$$

Putting everything together, we have:

$$C(t) = \frac{1}{2} \mathbf{c}^T S^{-1} \left(i\Omega + \coth\left(\frac{\beta}{2} D\right) \right) S^{-T} e^{tH} \mathbf{c} \quad (\text{B13})$$

and because of $S^{-1} \Omega S^{-T} = \Omega$ we can rewrite:

$$C(t) = \frac{1}{2} \mathbf{c}^T \left(i\Omega + S^{-1} \coth\left(\frac{\beta}{2} D\right) S^{-T} \right) e^{tH} \mathbf{c} \quad (\text{B14})$$

$$= \frac{1}{2} \mathbf{c}^T \left(i\Omega + S^{-1} \coth\left(\frac{\beta}{2} D\right) S^{-T} \right) S^T e^{tD\Omega} S^{-T} \mathbf{c} \quad (\text{B15})$$

$$= \frac{1}{2} \mathbf{c}^T S^{-1} \left(i\Omega + \coth\left(\frac{\beta}{2} D\right) \right) e^{tD\Omega} S^{-T} \mathbf{c} \quad (\text{B16})$$

$$(\text{B17})$$

Specifically for this kind of diagonal matrix D from the Williamson decomposition, where the diagonal entries come in pairs, we have $\Omega D \Omega = -D$ (equivalently $[\Omega, D] = 0$) such that:

$$e^{tD\Omega} = \sum_{n=0}^{\infty} \frac{t^n}{n!} (D\Omega)^n = \cos(Dt) + \sin(Dt)\Omega \quad (\text{B18})$$

Inserting this, we have:

$$C(t) = \frac{1}{2} \mathbf{c}^T S^{-1} \left(i\Omega + \coth\left(\frac{\beta}{2} D\right) \right) (\cos(Dt) + \sin(Dt)\Omega) S^{-T} \mathbf{c} \quad (\text{B19})$$

We may now realize that some of these terms are zero. To see this, we consider the transpose of a scalar, which must equal itself, and utilize the commutativity of Ω and D to show that it is equal to minus itself and thus equates to zero:

$$\mathbf{c}^T S^{-1} \Omega \cos(Dt) S^{-T} \mathbf{c} \stackrel{\text{transpose}}{=} \mathbf{c}^T S^{-1} \cos(Dt) (-\Omega) S^{-T} \mathbf{c} = 0 \quad (\text{B20})$$

$$\mathbf{c}^T S^{-1} \coth\left(\frac{\beta}{2} D\right) \sin(Dt) \Omega S^{-T} \mathbf{c} \stackrel{\text{transpose}}{=} \mathbf{c}^T S^{-1} (-\Omega) \sin(Dt) \coth\left(\frac{\beta}{2} D\right) S^{-T} \mathbf{c} = 0 \quad (\text{B21})$$

Together with $\Omega^2 = -\mathbb{1}$, we therefore end up with:

$$C(t) = \frac{1}{2} \mathbf{c}^T S^{-1} \left(-i \sin(Dt) + \coth\left(\frac{\beta}{2} D\right) \cos(Dt) \right) S^{-T} \mathbf{c} \quad (\text{B22})$$

We will now consider the noise spectrum, which is the Fourier transform of the bath correlation function:

$$S_\beta(\omega) = \int_{-\infty}^{\infty} C(t) e^{i\omega t} dt \quad (\text{B23})$$

By utilizing:

$$-i \int_{-\infty}^{\infty} \sin(Dt) e^{i\omega t} dt = \pi [\delta(\omega - D) - \delta(\omega + D)] \quad (\text{B24})$$

$$\int_{-\infty}^{\infty} \cos(Dt) e^{i\omega t} dt = \pi [\delta(\omega - D) + \delta(\omega + D)] \quad (\text{B25})$$

where scalars such as ω are understood to be proportional to the identity matrix $\omega \mathbb{1}$ and the delta function with a diagonal matrix as argument, is simply a diagonal matrix with elements $\delta(\omega - D)_{ii} = \delta(\omega - D_{ii})$, we recover:

$$S_\beta(\omega) = \frac{\pi}{2} \mathbf{c}^T S^{-1} \left[\delta(\omega - D) - \delta(\omega + D) + \coth\left(\frac{\beta}{2} D\right) [\delta(\omega - D) + \delta(\omega + D)] \right] S^{-T} \mathbf{c} \quad (\text{B26})$$

$$S_\beta(\omega) = \frac{\pi}{2} \mathbf{c}^T S^{-1} \left[\delta(\omega - D) - \delta(\omega + D) + \coth\left(\frac{\beta}{2} \omega\right) [\delta(\omega - D) - \delta(\omega + D)] \right] S^{-T} \mathbf{c} \quad (\text{B27})$$

where we have used that $\coth$ is an odd function. By defining the spectral density

$$J(\omega) = \frac{\pi}{2} \mathbf{c}^T S^{-1} \delta(\omega - D) S^{-T} \mathbf{c} \quad (\text{B28})$$

we can write the noise spectrum in the familiar form:

$$S_\beta(\omega) = \left(\coth\left(\frac{\beta}{2} \omega\right) + 1 \right) (J(\omega) - J(-\omega)) = \frac{2}{1 - e^{-\beta \omega}} (J(\omega) - J(-\omega)) \quad (\text{B29})$$

Exactly like Eq. A5, except, now we have a more general presentation of the spectral density.

We will now write the spectral density in a way that does not rely on the Williamson decomposition of the matrix H . Simply inserting $D = S^{-T} H S^{-1}$ does not perform the job. Instead, we will use the Sokhotski-Plemelj theorem to write the odd combination of delta functions as:

$$\pi [\delta(\omega - D) - \delta(\omega + D)] = \text{Im} \left[\frac{1}{\omega - D - i0^+} - \frac{1}{\omega + D - i0^+} \right] = \text{Im} \left[\frac{1}{\omega^2 - D^2 - \text{sgn}(\omega) i0^+} \right] D \quad (\text{B30})$$

We can then use that:

$$-D^2 = (-D)D = \Omega D \Omega D = \Omega S^{-T} H S^{-1} \Omega S^{-T} H S^{-1} = S \Omega H \Omega H S^{-1} \quad (\text{B31})$$

Thus, we can write the odd-combination of the spectral density $J^o(\omega)$ as:

$$J^o(\omega) := J(\omega) - J(-\omega) = \mathbf{c}^T S^{-1} S \text{Im} \left[\frac{1}{\omega^2 + (\Omega H)^2 - \text{sgn}(\omega) i0^+} \right] S^{-1} D S^{-T} \mathbf{c} \quad (\text{B32})$$

Lastly, $S^{-1} D S^{-T} = -S^{-1} \Omega D \Omega S^{-T} = -\Omega H \Omega$ and we arrive at:

$$J^o(\omega) = -\mathbf{c}^T \text{Im} \left[\frac{1}{\omega^2 + (\Omega H)^2 - \text{sgn}(\omega) i0^+} \right] \Omega H \Omega \mathbf{c} \quad (\text{B33})$$

The problem of finding the spectral density has now been turned into a matrix inversion problem. The practicality of this becomes apparent in the later examples.

We may simplify the expression further if the examples we consider are trivial in the momentum sector and only couple in the position sector. For example, in the basis $\mathbf{r} = (\hat{x}_1, \hat{x}_2, \dots, \hat{p}_1, \hat{p}_2, \dots)^T$, we consider:

$$\Omega = \begin{pmatrix} 0 & \mathbb{1} \\ -\mathbb{1} & 0 \end{pmatrix} \quad H = \begin{pmatrix} X & 0 \\ 0 & \mathbb{1} \end{pmatrix} \quad \mathbf{c} = \begin{pmatrix} \mathbf{c}_x \\ 0 \end{pmatrix} \quad (\text{B34})$$

Then $(\Omega H)^2 = \begin{pmatrix} -X & 0 \\ 0 & -X \end{pmatrix}$ and $\Omega H \Omega = -\begin{pmatrix} \mathbb{1} & 0 \\ 0 & X \end{pmatrix}$ which results in:

$$J^o(\omega) = \mathbf{c}_x^T \text{Im} \left[\frac{1}{\omega^2 - X - \text{sgn}(\omega)i0^+} \right] \mathbf{c}_x \quad (\text{B35})$$

1. Spectral density of interacting lossy harmonic modes

Consider N bosons that may couple directly to the system, as described by a vector $\mathbf{c}$, and interact among themselves through coupling constants contained in a matrix we call χ . In addition, each boson may be coupled to its own structured bath. Such a scenario is visualized in Fig. 10. For clarity, let us write out the Hamiltonian we have in

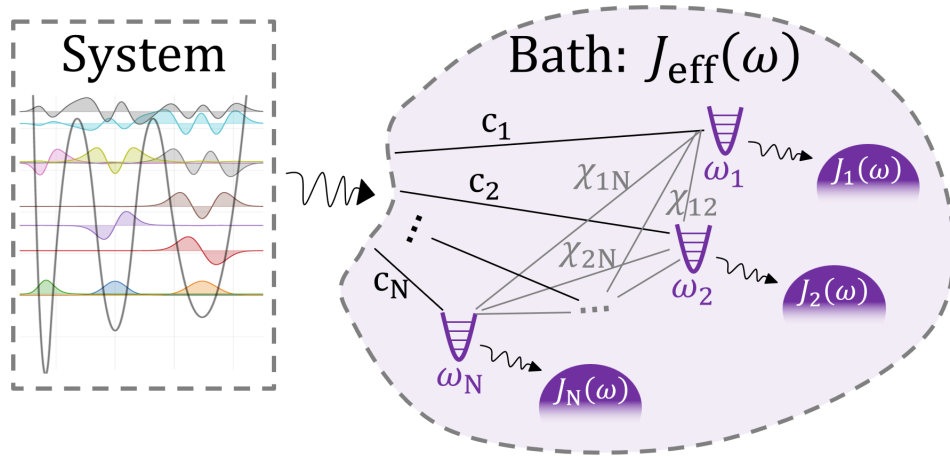


FIG. 10. Schematic of the efficient modeling of an interacting harmonic environment in terms of an effective spectral density for the full environment (Eq. B39). The example used for 'the system' is the triple well PES from sec. IV A 3.

mind explicitly:

$$H = H_R + \sum_n \frac{c_n^2}{2\omega_n^2} \hat{R}^2 + \hat{R} \sum_n c_n \hat{x}_n + \sum_n \left[\frac{\hat{p}_n^2}{2} + \omega_n^2 \frac{\hat{x}_n^2}{2} \right] + \sum_{n \neq m} \frac{\chi_{nm}}{2} \hat{x}_n \hat{x}_m + \sum_n H_{\text{CL}}^{(n)} \quad (\text{B36})$$

H_R is the system Hamiltonian and $\hat{R}$ is the system operator that couples to the harmonic environment. The mass-weighted coordinates of the harmonic environment $\hat{x}_n$, are mutually interacting and they each experience their own Harmonic environment through the inclusion of the $H_{\text{CL}}^{(n)}$

$$H_{\text{CL}}^{(n)} = \sum_l \frac{\hat{p}_{n,l}^2}{2} + \frac{\omega_{n,l}^2}{2} \left(\hat{q}_{n,l} - \frac{c_{n,l}}{\omega_{n,l}^2} \hat{x}_n \right)^2 \quad (\text{B37})$$

which gives rise to the spectral densities $J_n(\omega) = \frac{\pi}{2} \sum_l \frac{c_{n,l}^2}{\omega_{n,l}} \delta(\omega - \omega_{n,l})$ that characterize the local harmonic environment for the n 'th oscillator $\hat{x}_n$.

This Hamiltonian (Eq. B36), with formally infinitely many degrees of freedom, is of the form Eq. B1. In the appropriate basis, one can then write the coupling matrix of the full harmonic environment, X (eq. B34) in the following block form:

$$X = \begin{pmatrix} \chi & C^T \\ C & W \end{pmatrix} \quad (\text{B38})$$

where W is diagonal and contains all the frequencies of the $J_i(\omega)$ dofs, C is a matrix with N columns that couples each boson i to its respective bath $J_i(\omega)$ and χ is the general symmetric $N \times N$ matrix that couples the lossy bosons. The system only couples to the χ part, such that Eq. B35 becomes:

$$J_{\text{eff}}(\omega) = \text{Im} \left(\begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}^T \begin{pmatrix} \omega^2 - \omega_1^2 - iJ_1^o(\omega) - \tilde{\mathcal{R}}_1(\omega) & -\chi_{12} & \dots & -\chi_{1N} \\ -\chi_{21} & \omega^2 - \omega_2^2 - iJ_2^o(\omega) - \tilde{\mathcal{R}}_2(\omega) & \dots & -\chi_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ -\chi_{N1} & -\chi_{N2} & \dots & \omega^2 - \omega_N^2 - iJ_N^o(\omega) - \tilde{\mathcal{R}}_N(\omega) \end{pmatrix}^{-1} \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix} \right) \quad (\text{B39})$$

on account of the inversion of a block matrix $\begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} (A - BD^{-1}C)^{-1} & \dots \\ \dots & \dots \end{pmatrix}$. The Schur complement produced entries along the diagonal, which we rewrote by again applying the Sokhotski-Plemelj theorem:

$$\sum_j c_{n,j}^2 \frac{1}{\omega^2 - \omega_{n,j}^2 - \text{sgn}(\omega) i 0^+} = \sum_j c_{n,j}^2 \left[\text{sgn}(\omega) i \pi \delta(\omega^2 - \omega_{n,j}^2) + \mathcal{P} \left(\frac{1}{\omega^2 - \omega_{n,j}^2} \right) \right] = iJ_n^o(\omega) + \mathcal{R}_n(\omega) \quad (\text{B40})$$

We applied the identity: $\text{sgn}(\omega) \delta(\omega^2 - \nu^2) = \frac{1}{2\omega} [\delta(\omega - \nu) - \delta(\omega + \nu)]$ (for $\nu > 0$) and defined the principal value integral:

$$\mathcal{R}_n(\omega) = \frac{2}{\pi} \mathcal{P} \int_0^\infty \frac{\omega' J_n(\omega')}{\omega^2 - \omega'^2} d\omega' \quad (\text{B41})$$

$$= \mathcal{P} \int_{-\infty}^\infty \frac{d\omega'}{\pi} \frac{\omega' J_n^o(\omega')}{\omega^2 - \omega'^2} \quad (\text{B42})$$

$$= \mathcal{P} \int_{-\infty}^\infty \frac{d\omega'}{2\pi} J_n^o(\omega') \left(\frac{1}{\omega - \omega'} - \frac{1}{\omega + \omega'} \right) \quad (\text{B43})$$

$$= \mathcal{P} \int_{-\infty}^\infty \frac{d\omega'}{\pi} \frac{J_n^o(\omega')}{\omega - \omega'} \quad (\text{B44})$$

Lastly, we reintroduced the neglected reorganization energies by the following consideration:

$$\hat{H} = \dots + \frac{\hat{p}_n^2}{2} + \frac{\omega_n^2}{2} \hat{x}_n^2 + \hat{x}_n^2 \underbrace{\sum_j \frac{c_{n,j}^2}{2\omega_{n,j}^2}}_{\int_0^\infty \frac{d\omega}{\pi} \frac{J_n(\omega)}{\omega}} + \dots \Rightarrow \omega_n^2 \rightarrow \omega_n^2 + 2 \int_0^\infty \frac{d\omega}{\pi} \frac{J_n(\omega)}{\omega} \quad (\text{B45})$$

which means that we may incorporate them into $\mathcal{R}_n(\omega)$ in the following way:

$$\tilde{\mathcal{R}}_n(\omega) = \mathcal{R}_n(\omega) + 2 \int_0^\infty \frac{d\omega'}{\pi} \frac{J_n(\omega')}{\omega'} = \omega \mathcal{P} \int_{-\infty}^\infty \frac{d\omega'}{\pi} \frac{J_n^o(\omega')}{(\omega - \omega') \omega'} \quad (\text{B46})$$

It will be a helpful reminder, that if the pole structure of the bath spectral density is simple e.g. if $J_n^o(\omega) = \frac{f(\omega)}{g(\omega)}$ where $g(\omega)$ accounts for the simple poles, then:

$$\tilde{\mathcal{R}}(\omega) = -2\omega \text{Im} \left(\sum_{\omega_p} \frac{f(\omega_p)}{(\omega - \omega_p) \omega_p g'(\omega_p)} \right) \quad (\text{B47})$$

where ω_p are the zeros of $g(\omega)$ in the upper-half of the complex plane. For the Drude-Lorentz spectral density

$$J^{(DL)}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2} \quad \tilde{\mathcal{R}}^{(DL)}(\omega) = \frac{2\lambda\omega^2}{\omega^2 + \gamma^2} \quad (\text{B48})$$

For the following section, and in the main text in general, we use $J(\omega)$ and its odd-version $J^o(\omega) = J(\omega) - J(-\omega)$ interchangeably.

2. Spectral density of interacting cavity and spectator mode, averaged over molecule orientation

We consider the spectral density in Eq. 7 while ignoring the phonon bath:

$$J(\omega) = \text{Im} \frac{\Gamma_c(\omega)c_Q^2 + \eta_R^2\Gamma_Q(\omega)2\omega_c^3\eta_c^2 + 4\eta_Q\eta_R(\omega^2 - i\alpha\omega)c_Q\omega_c\eta_c^2 + 4\eta_Q^2\eta_R^2(\omega^2 - i\alpha\omega)\omega_c^2\eta_c^4}{\Gamma_c(\omega)\Gamma_Q(\omega) - 2(\omega^2 - i\alpha\omega)\omega_c\eta_c^2\eta_Q^2} \quad (\text{B49})$$

where $\Gamma_c(\omega) = \omega^2 - \omega_c^2 - i\alpha\omega$ and $\Gamma_Q(\omega) = \omega^2 - \omega_Q^2 - \left(i + \frac{\omega}{\gamma_Q}\right) J_Q^{(DL)}(\omega)$

We will now introduce $\eta_R = \tilde{\eta}_R \cos(\theta - \phi)$ and $\eta_Q = \tilde{\eta}_Q \cos(\phi)$ where θ is the angle between the reaction-coordinate and the spectator mode and ϕ is the angle between the cavity polarization and the spectator mode dipole moment. We are interested in averaging over ϕ

$$\langle J(\omega) \rangle_\phi = \frac{1}{\pi} \int_0^\pi J(\omega) d\phi \quad (\text{B50})$$

We apply $\cos(\theta - \phi) = \cos(\theta)\cos(\phi) + \sin(\theta)\sin(\phi)$ and utilize that the average over odd functions equals zero which together with polynomial division gives:

$$\left\langle \frac{1}{a - b \cos^2(\phi)} \right\rangle_\phi = \frac{1}{\sqrt{a(a-b)}} \quad (\text{B51})$$

$$\left\langle \frac{\eta_R^2}{a - b \cos^2(\phi)} \right\rangle_\phi = \tilde{\eta}_R^2 \left(\frac{\sin^2(\theta)}{\sqrt{a(a-b)}} + \frac{\cos(2\theta)}{b} \left(\sqrt{\frac{a}{a-b}} - 1 \right) \right) \quad (\text{B52})$$

$$\left\langle \frac{\eta_R\eta_Q}{a - b \cos^2(\phi)} \right\rangle_\phi = \tilde{\eta}_R\tilde{\eta}_Q \frac{\cos(\theta)}{b} \left(\sqrt{\frac{a}{a-b}} - 1 \right) \quad (\text{B53})$$

$$\left\langle \frac{\eta_R^2\eta_Q^2}{a - b \cos^2(\phi)} \right\rangle_\phi = \tilde{\eta}_R^2\tilde{\eta}_Q^2 \left(\frac{\sin^2(\theta)}{b} \left(\sqrt{\frac{a}{a-b}} - 1 \right) + \cos(2\theta) \left(\frac{a}{b^2} \left(\sqrt{\frac{a}{a-b}} - 1 \right) - \frac{1}{2b} \right) \right) \quad (\text{B54})$$

with $a = \Gamma_c(\omega)\Gamma_Q(\omega)$ and $b = 2\tilde{\eta}_Q^2(\omega^2 - i\alpha\omega)\omega_c\eta_c^2$ where $\cos^2(\theta) - \sin^2(\theta) = \cos(2\theta)$ was also used.

Supplemental Material C: Reaction Coordinate Potentials & vibrational state details

We consider a model system which consists of a reaction coordinate on a one dimensional potential energy surface.

$$H_S = \frac{\hat{P}^2}{2M} + V(\hat{X}) \quad (\text{C1})$$

1. Double Well

One way to parameterize the symmetric double well potential is as a quartic: $V_{\text{sym}}(x) = E_b \left(1 - \left(\frac{\hat{R}}{R_0} \right)^2 \right)^2$ However, we are interested in a parameterization where we have control of the product-, reactant- and barrier frequencies independently. To this end, we consider a piecewise function of quadratic polynomials.

$$V(x) = \begin{cases} \frac{1}{2}M\omega_l^2(x + x_0^l)^2 - E_l & x < x_l \\ E_b - \frac{1}{2}M\omega_0^2x^2 & x_l \leq x \leq x_r \\ \frac{1}{2}M\omega_r^2(x + x_0^r)^2 - E_r & x_r < x \end{cases} \quad (\text{C2})$$

where we only have to specify the frequencies and offsets, with the location of the parabolas and the regions given by:

$$x_0^l = \sqrt{\frac{2(\omega_0^2 + \omega_l^2)(E_0 + E_l)}{M\omega_0^2\omega_l^2}} \quad (\text{C3})$$

$$x_l = -\frac{\omega_l^2}{\omega_0^2 + \omega_l^2} x_0^l \quad (\text{C4})$$

$$x_0^r = -\sqrt{\frac{2(\omega_0^2 + \omega_r^2)(E_0 + E_r)}{M\omega_0^2\omega_r^2}} \quad (\text{C5})$$

$$x_r = -\frac{\omega_r^2}{\omega_0^2 + \omega_r^2} x_0^r \quad (\text{C6})$$

$$(\text{C7})$$

a. Derivation

We require any contiguous pair of parabolas to touch at a single point:

$$a(x+b)^2 - c = C - A(x+B)^2 \quad (\text{C8})$$

$$\Rightarrow 0 = (a+A)x^2 + 2(ab+AB)x + ab^2 + AB^2 - c - C \quad (\text{C9})$$

$$(\text{C10})$$

Where a,b,c and A,B,C denote the parameters of the well and barrier, respectively. We require a unique solution for x , which is the case when the discriminant is zero (for real parameters):

$$x = -\frac{ab+AB}{a+A} \Leftrightarrow (ab+AB)^2 = (a+A)(ab^2 + AB^2 - c - C) \quad (\text{C11})$$

Furthermore, we would like to solve for either b or B since the other parameters are fixed. To this end, the rightmost identity can be rewritten as:

$$(b-B)^2 = \frac{(a+A)(c+C)}{aA} \quad (\text{C12})$$

Associating a,b,c and A,B,C with the parameters of the potential results in the expressions above.

2. Triple Well

In order to study competing pathways, we consider a potential energy surface with three wells. We can utilize the same parameterization as above, which results in:

$$V(x) = \begin{cases} \frac{1}{2}M\omega_{ll}^2(x+x_0^l)^2 - E_{ll} & -\infty < x \leq x_{ll} \\ E_l - \frac{1}{2}M\omega_l^2(x+x_0^l)^2 & x_{ll} < x \leq x_l \\ \frac{1}{2}M\omega_0^2x^2 - E_0 & x_l < x \leq x_r \\ E_r - \frac{1}{2}M\omega_r^2(x+x_0^r)^2 & x_r < x \leq x_{rr} \\ \frac{1}{2}M\omega_{rr}^2(x+x_0^r)^2 - E_{rr} & x_{rr} < x < \infty \end{cases} \quad (\text{C13})$$

with

$$x_0^l = \sqrt{\frac{2(\omega_0^2 + \omega_l^2)(E_0 + E_l)}{M\omega_0^2\omega_l^2}} \quad (\text{C14})$$

$$x_l = -\frac{\omega_l^2}{\omega_0^2 + \omega_l^2} x_0^l \quad (\text{C15})$$

$$x_0^l = x_0^l + \sqrt{\frac{2(\omega_{ll}^2 + \omega_l^2)(E_{ll} + E_l)}{M\omega_{ll}^2\omega_l^2}} \quad (\text{C16})$$

$$x_{ll} = -\frac{\omega_l^2 x_0^l + \omega_{ll}^2 x_0^l}{\omega_l^2 + \omega_{ll}^2} \quad (\text{C17})$$

$$x_0^r = -\sqrt{\frac{2(\omega_0^2 + \omega_r^2)(E_0 + E_r)}{M\omega_0^2\omega_r^2}} \quad (\text{C18})$$

$$x_r = -\frac{\omega_r^2}{\omega_0^2 + \omega_r^2} x_0^r \quad (\text{C19})$$

$$x_0^{rr} = x_0^r - \sqrt{\frac{2(\omega_{rr}^2 + \omega_r^2)(E_{rr} + E_r)}{M\omega_{rr}^2\omega_r^2}} \quad (\text{C20})$$

$$x_{rr} = -\frac{\omega_r^2 x_0^r + \omega_{rr}^2 x_0^{rr}}{\omega_r^2 + \omega_{rr}^2} \quad (\text{C21})$$

The ground state of the different wells can then be made approximately degenerate, by specifying the offset energy as $E = \frac{\omega_{\text{well}}}{2}$. This was done for triple well potential in Figure 7a, with some additional fine-tuning to get the excited-state degeneracy seen there. Alternatively, which was done for the double wells in figure 5, one can instead specify $E_{l,r} = 0$ and add a linear term $\frac{1}{2} \frac{\omega_l - \omega_r}{x_l + x_r} x$ to the potential. Probably, these approaches are equivalent even though we do not attempt to show that here.

3. Localized states

In order to analyze the dynamics in terms of intra- and inter-well dynamics, we consider localized states which are not eigenstates of the reaction coordinate Hamiltonian $H_R = \sum_n E_n |n\rangle \langle n|$. We obtain these localized states by diagonalizing the reaction coordinate operator in the sub-space of the relevant delocalized energy-states, e.g.:

$$\begin{pmatrix} \langle n_1 | \hat{R} | n_1 \rangle & \langle n_1 | \hat{R} | n_2 \rangle & \langle n_1 | \hat{R} | n_3 \rangle \\ \langle n_2 | \hat{R} | n_1 \rangle & \langle n_2 | \hat{R} | n_2 \rangle & \langle n_2 | \hat{R} | n_3 \rangle \\ \langle n_3 | \hat{R} | n_1 \rangle & \langle n_3 | \hat{R} | n_2 \rangle & \langle n_3 | \hat{R} | n_3 \rangle \end{pmatrix} = O \overbrace{D}^{\text{diag}} O^T \quad (\text{C22})$$

The localized states $|l_{1,2,3}\rangle$ are then given by:

$$|l_i\rangle = \sum_{j=1,2,3} O_{ji} |n_j\rangle \quad (\text{C23})$$

In figure 4, the two ground states and the two excited states were hybridized to produce the localized states shown. In figure 5, even though the eigenstates were already approximately localized due to the broken symmetry of the potentials, we still hybridized the ground states and also the two states closest to Frequency = 1200cm⁻¹ to obtain the localized states shown. In figure 7a the three excited states around 1570cm⁻¹ were localized with this process.

4. Numerics

For the simulations, the reaction coordinate was expressed in the eigen basis of the system Hamiltonian (without the reorganization term). The system Hamiltonian was numerically diagonalized after expressing it in the Discrete Variable Representation (DVR) based on the sinc-function⁷⁷. Sinc-DVR resembles that of discretizing the position basis at N points spaced by Δx , where operators that are a function of $\hat{X}$ remain diagonal in this representation.

However, instead of a finite-difference representation of the momentum operator, in sinc-DVR, the momentum operator is instead represented as:

$$(\hat{P})_{nm} = \begin{cases} 0 & n = m \\ \frac{-i(-1)^{n-m}}{\Delta x(n-m)} & n \neq m \end{cases} \quad (\text{C24})$$

$$(\hat{P}^2)_{nm} = \begin{cases} \frac{\pi^2}{3\Delta x^2} & n = m \\ \frac{2(-1)^{n-m}}{\Delta x^2(n-m)^2} & n \neq m \end{cases} \quad (\text{C25})$$

N and Δx was chosen such that convergence was observed in the eigenvalue of the relevant states.

Supplemental Material D: Details on Hierarchical Equations Of Motion

We refer to our choice of method for simulating the dynamics as HEOM⁴⁴, but instead of viewing the equations of motion as a hierarchy of coupled differential equations, we treat it as the system coupled to a collection of bosons. See e.g. references^{45–49}. For example, the master equation^{46,48,49}:

$$\begin{aligned} \frac{d\rho}{dt} = \mathcal{L}\rho = -i & \left[\hat{H}_R + \sum_k \text{Im}(z_k) \hat{a}_k^\dagger \hat{a}_k + \sum_k \text{Re}(\sqrt{d_k}) \hat{R} (\hat{a}_k + \hat{a}_k^\dagger), \rho \right] \\ & + 2 \sum_k \text{Re}(z_k) \left(\hat{a}_k \rho \hat{a}_k^\dagger - \frac{1}{2} \{ \hat{a}_k^\dagger \hat{a}_k, \rho \} \right) \\ & - 2 \sum_k \text{Im}(\sqrt{d_k}) \left(\hat{a}_k \rho \hat{R} - \frac{1}{2} \{ \hat{R} \hat{a}_k, \rho \} \right) \\ & - 2 \sum_k \text{Im}(\sqrt{d_k}) \left(\hat{R} \rho \hat{a}_k^\dagger - \frac{1}{2} \{ \hat{a}_k^\dagger \hat{R}, \rho \} \right) \end{aligned} \quad (\text{D1})$$

constitutes an open quantum system for the system S with a harmonic environment of K bosons with BCF:

$$C(t) = \sum_k^K d_k e^{-z_k t} \quad (\text{D2})$$

provided that the initial state factorizes as $\rho(t=0) = \rho_S(0) \otimes^K |0\rangle\langle 0|$. The noise spectrum is consequently:

$$S_\beta(\omega) = \sum_k^K 2 \frac{\text{Re}(d_k z_k^*) - \text{Im}(d_k) \omega}{\text{Re}(z_k)^2 + (\omega - \text{Im}(z_k))^2} \quad (\text{D3})$$

In order to better understand the generator in Eq. D1, consider $\text{Im}(d_k) = 0$. In such cases, the structure is equivalent to a Lindbladian (a generator for a completely positive, trace preserving dynamical semigroup). This would constitute a physical setup, where the system is coupled to bosons with Markovian loss and the noise spectrum is decomposed as a sum of Lorentzians. However, in general, this will not be an efficient decomposition and K would need to be too large to be practical. Allowing $\text{Im}(d_k) \neq 0$ alleviates this problem, since the terms in Eq. D3 are now able to acquire negative values and may cancel each other in a manner that assists the decomposition⁴⁷. Allowing this comes at the cost of the two terms in Eq. D1, which breaks the complete positivity of the generator. This means that, even though Eq. D1 is trace-preserving ($\text{Tr}[\rho] = 0$) the full density matrix $\rho(t)$ may acquire negative eigenvalues and no longer corresponds to a physical setup. However, crucially, the reduced density matrix $\rho_S(t) = \text{Tr}_K[\rho(t)]$ is still a physical density matrix.^{46,48,49}

Ref.⁷⁸ pointed to a potential problem. Namely, when cutoffs of the auxiliary modes are inevitably introduced for numerical implementation, the evolution can diverge in a way that is seemingly not just remedied by increasing the cutoff. However, ref.⁴⁶ notes that the presence of the divergence depends on the choice of 'gauge' when decomposing $C(t-t')$. This gauge-freedom is made manifest in the master equation by applying a similarity transformation using $\kappa_k \hat{a}_k^\dagger \hat{a}_k$, which does not change the initial and final conditions for the evolution, but simply transforms $\hat{a}_k \rightarrow \hat{a}_k / \kappa_k$ and $\hat{a}_k^\dagger \rightarrow \kappa_k \hat{a}_k^\dagger$. The κ_k 's can then be chosen such that norm of the CP-breaking term is minimized (but non-zero), which,

in that case, remedied the divergence. In the case of a single system operator interacting with the environment, i.e. a scalar valued BCF, the optimal gauge is known and Eq. D1 is written in this gauge. The ubiquitous and successful application of these CP-breaking formulations suggests that these concerns are not dire and also, in our experience, Eq. D1 has yet to exhibit divergent behavior.

Nonetheless, a lack in understanding how and when such divergent behavior arises and because it may be related to the CP-breaking of these types of methods, could motivate the search for Markovian embeddings that preserves complete-positivity. There are already numerous formulations that achieve this (see refs. ^{51,52,79–81} to name a few). However, an overview of the drawbacks for each and every method, such as computational complexity and robustness, has yet to be developed. Thus, the tried and tested HEOM formulations remain our choice of method.

Another useful version of HEOM is obtained from Eq. D1 by applying a similarity transformation under $e^{\hat{a}_k \bullet \hat{a}_k^\dagger}$, which has the following effects^{48,49}:

$$\rho_S(t) = \text{Tr}_K [\rho(t)] \rightarrow \langle \mathbf{0} | \rho(t) | \mathbf{0} \rangle \quad (\text{D4})$$

$$\hat{a}_k^\dagger \bullet \rightarrow \hat{a}_k^\dagger \bullet + \bullet \hat{a}_k^\dagger \quad (\text{D5})$$

$$\bullet \hat{a}_k^\dagger \rightarrow \hat{a}_k^\dagger \bullet + \bullet \hat{a}_k^\dagger \quad (\text{D6})$$

$$\hat{a}_k \rightarrow \hat{a}_k \quad (\text{D7})$$

Now, the reduced density matrix for the system can be obtained by simply projecting the auxiliary modes onto the vacuum^{47–49}. This is helpful for both the hierarchical approaches, where you can now read-off the density matrix from the lowest tier and also tensor networks, where it allows you to place left and right dofs of the auxiliary modes independently of each other, since the trace could otherwise introduce loops that in the worst case could be difficult to contract. The price for this benefit, is that the master equation:

$$\dot{\rho}(t) = -i \left[\hat{H}_S, \rho(t) \right] + \sum_k^K \left\{ -z_k \hat{a}_k^\dagger \hat{a}_k \rho(t) - i\sqrt{d_k} \left(\hat{a}_k + \hat{a}_k^\dagger \right) \hat{S} \rho(t) - i\sqrt{d_k^*} \hat{S} \rho(t) \hat{a}_k^\dagger + H.C. \right\} \quad (\text{D8})$$

Is no longer trace perserving:

$$\text{Tr} [\dot{\rho}(t)] = \sum_k -2 \text{Re}(z_k) \text{Tr} \left[\hat{a}_k^\dagger \hat{a}_k \rho(t) \right] - i \text{Tr} \left[\left(\sqrt{d_k} \hat{a}_k^\dagger - \sqrt{d_k^*} \hat{a}_k \right) \hat{S} \rho(t) \right] \quad (\text{D9})$$

For weak to moderate coupling, this does not pose a problem and on the contrary, we experienced that simulating the dynamics with Eq. D8 required fewer computational resources than Eq. D1 (e.g. due to a lower bond-dimension). However, we did not carry out a precise comparison and it remains anecdotal information. Nonetheless, for strong coupling scenarios the trace can grow uncontrollably and one should resort to the trace preserving formulation, Eq. D1.

1. Decomposing the Bath Correlation Function

We will now describe how to obtain the parameters d_k and z_k . There exists multiple methods to fit a given bath correlation function by a sum of complex exponentials. We rely on the AAA-algorithm^{57,58} to efficiently acquire the decomposition in Eq. 22. In essence, $S_\beta(\omega)$ is decomposed as a sum of simple poles p_n with residues r_n within a window of frequencies $\omega \in [\omega_l, \omega_h]$, such that:

$$d_k = -i\omega_- r_k \quad z_k = i(\omega_- p_k + \omega_+) \quad (\text{D10})$$

where $\omega_\pm = (\omega_h \pm \omega_l)/2$ and k labels the poles in the lower half of the complex plane. The decomposition can usually be carried out to numerical accuracy with K on the order of 10 – 100, depending on the intricacies of the noise spectrum.

The reorganization energy in terms of the effective spectral density can be written as:

$$\lambda = \frac{1}{\pi} \int_0^\infty d\omega \frac{J(\omega)}{\omega} \quad (\text{D11})$$

$$= \int_0^\infty \frac{d\omega}{2\pi} (1 - e^{-\beta\omega}) \frac{S(\omega)}{\omega} \quad (\text{D12})$$

$$= \int_{-\infty}^\infty \frac{d\omega}{2\pi} \frac{S(\omega)}{\omega} \quad (\text{D13})$$

$$= \int_{-\infty}^\infty \frac{d\omega}{2\pi} \int_{-\infty}^\infty dt C(t) \frac{e^{i\omega t}}{\omega} \quad (\text{D14})$$

$$= \frac{i}{2} \int_{-\infty}^\infty C(t) \text{sgn}(t) \quad (\text{D15})$$

$$= - \int_0^\infty \text{Im}[C(t)] dt \quad (\text{D16})$$

by using that $S(\omega) = \frac{2}{1-e^{-\beta\omega}} J(\omega) = e^{\beta\omega} S(-\omega) = \int_{-\infty}^\infty dt C(t) e^{i\omega t}$ and $C(t) = C^*(-t)$. Then the reorganization energy for our emulating bath is easily obtained as Eq. 19.

2. Effective Markovian correction

As detailed in sec. D, the noise spectrum is decomposed into contributions from each auxiliary mode. We have found that some of these auxiliary modes constitute a Markovian bath. Thus, instead of treating these degrees of freedom explicitly, we can incorporate them as a jump operator on the system degree of freedom. This can be made precise using the Universal Lindblad Equation⁵⁹. Their rigorous analysis proves that a Gaussian environment constitutes a Markovian bath once the characteristic correlation time of the bath τ remain small compared to the rate of bath-induced evolution Γ . Once $\Gamma\tau \ll 1$ the Markovian dynamics are accurately described by introducing the jump operator $L = \sum_{n,m} \sqrt{S_\beta(E_m - E_n)} S_{nm} |n\rangle\langle m|$ for the system. Ref.⁵⁹ also provide an expression for the Hermitian Lamb shift, which for Markovian noise spectrums (e.g. $S_\beta(\omega) \sim \text{constant}$) resemble the reorganization term:

$$H_{\text{Lamb Shift}} = \sum_{n,m,l} \int_{-\infty}^\infty \frac{d\omega}{2\pi} \frac{\sqrt{S_\beta(\omega - (E_n - E_l))} S_\beta(\omega + E_m - E_l)}{\omega} S_{nl} S_{lm} |n\rangle\langle m| \approx \int_{-\infty}^\infty \frac{d\omega}{2\pi} \frac{S_\beta(\omega)}{\omega} \hat{S}^2 = \lambda \hat{S}^2 \quad (\text{D17})$$

So the Markovian bath contributes both a reorganization term and a Lamb shift that, seemingly, should be added together. However, the system is formally in a product state with the Markovian environment and thus no renormalization of the system from hybridizing with the Markovian bath is expected and instead the two terms should cancel rather than add⁸². Thus, for the auxiliary modes that are deemed Markovian, we simply remove their contribution to the reorganization energy. Also, E_n and $|n\rangle$ actually refer to the energies and eigenstates of the system dressed by the residual environment, which may be difficult to compute exactly. As they note, however, it can be efficiently computed perturbatively and for our intents and purposes, we have found simply the system Hamiltonian plus the reorganization term $H_R + \lambda \hat{R}^2 = \sum_n E_n |n\rangle\langle n|$ to work well.

While ref.⁵⁹ controls the Markovian approximation in terms of $\Gamma\tau$, they also define $\Gamma_0\tau_0 = 4 \int_0^\infty t |C(t)| dt$ which we found more practical to work with than $\Gamma\tau$. While $\Gamma_0\tau_0 \leq \Gamma\tau$, ref.⁸³ use $2\Gamma_0\tau_0$ to control their related approximation. Thus, we can rigorously control the Markovian approximation of the auxiliary modes in terms of:

$$2\Gamma_0\tau_0 = 8 \int_0^\infty t |C(t)| dt = 8 \int_0^\infty t \left| \sum_k d_k e^{-z_k t} \right| dt \leq 8 \sum_k \int_0^\infty t |d_k e^{-z_k t}| dt = 8 \sum_k \frac{|d_k|}{\text{Re}(z_k)^2} \quad (\text{D18})$$

Interestingly, for a single exponential with a real prefactor $C(t) = |d|e^{-zt}$, we can compute $\Gamma\tau = \frac{2}{\pi} 8 \frac{|d|}{\text{Re}(z)^2}$ (using integrals of the zeroth modified Bessel function of the second kind) which compared to $2\Gamma_0\tau_0$ is smaller by a factor $\frac{2}{\pi}$, in agreement with $2\Gamma_0\tau_0$ being a good way to bound the error.

In Figure 11 we exemplify this approach by simulating the symmetric double well interacting with the phonon bath, for different values of $2\Gamma_0\tau_0$. The phonon bath has been fitted to high-accuracy and the number of auxiliary modes is $|K| = 19$. However, many of these modes basically constitute a Markovian environment, so choosing M of these such that $2\Gamma_0\tau_0 \approx 0.01$, reduces the number of auxiliary modes that need to be simulated explicitly to $|K| = |K \setminus M| = 2$.

That is, the evolution induced by $|M| = 17$ of these auxiliary modes can be captured by the system jump operator L , as seen by the resulting reaction rate $k(t) = \frac{\dot{P}_r(t)}{1-2P_r(t)}$ (Eq. E3). The resulting evolution is nearly indistinguishable from the case where all the auxiliary modes have been simulated explicitly, $2\Gamma_0\tau_0 = 0$ and $|\mathcal{K}| = 19$. However, increasing $|M| = 18$ and thus $2\Gamma_0\tau_0 \approx 0.06$, begins to introduce a deviation on the calculated reaction rate k . Based on this, we have chosen to truncate K to $\mathcal{K}$ by choosing M such that $2\Gamma_0\tau_0$ is at most 0.01 (in practice often smaller) for all our simulations. This allows us to substantially reduce the number of explicit auxiliary modes while staying true to the dynamics. Performing a less accurate

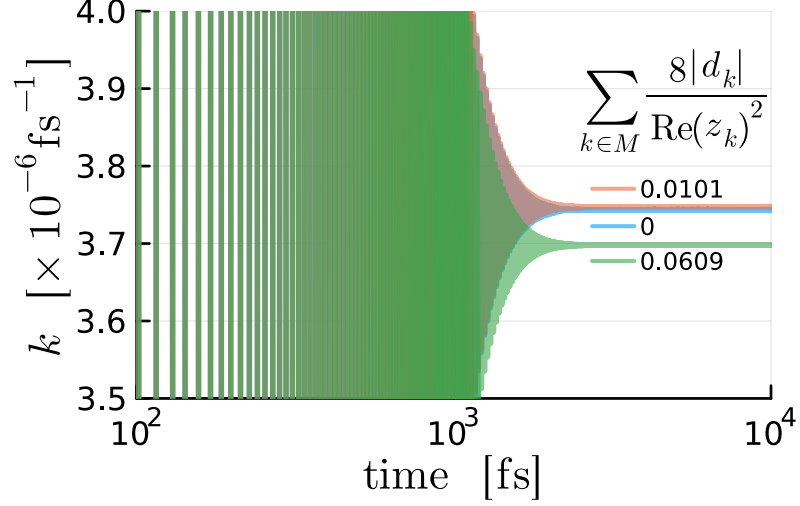


FIG. 11. Comparison of $k(t) = \frac{\dot{P}_r(t)}{1-2P_r(t)}$ (Eq. E3) for different levels of Markovian compression of the phonon bath. The observed oscillations correspond to the inherent time scale of the phonon bath $\gamma_R^{-1} \approx \beta \approx 25\text{fs}$ and originates from the quenched dynamics since the system is initialized in a product state with the phonon bath. The simulations were carried out using tensor networks, and for this environment, the number of physical indices needed to represent the environment could be reduced from $2 \times 19 = 38$ to $2 \times 2 = 4$ without having any effect on the observed reaction rate. Compare this to reducing the number of indices by performing a less accurate decomposition of the bath correlation function, which was found to result in a large deviation in the observed rate.

Supplemental Material E: Reaction Rate Analysis

We define the reaction rate constant by following e.g. references^{37,38,84} For the double well potential, we define the left well as the reactant region with partial concentration P_r and the right well as the product region with partial concentration $P_p = 1 - P_r$.

P_r is defined as an expectation value of the projection onto the left well:

$$P_r(t) = \text{Tr} [\hat{P}_r \rho(t)] \quad \text{with} \quad \hat{P}_r = \int_{-\infty}^0 dR |R\rangle \langle R| \quad (\text{E1})$$

Past some transient behavior, we can assume the expectation values follow the rate equation:

$$\dot{P}_r(t) = -kP_r(t) + k_bP_p(t) \quad (\text{E2})$$

where k is the forward reaction rate and k_b is the backward reaction rate, which determine the equilibrium distribution $\frac{k_b}{k} = \frac{P_r(t \rightarrow \infty)}{P_p(t \rightarrow \infty)}$. Using this, one can isolate k in Eq. E2:

$$k = \frac{\dot{P}_r(t)}{-P_r(t) + \frac{P_r^{eq}}{P_p^{eq}}(1 - P_r(t))} \quad (\text{E3})$$

The time derivative of $P_r(t)$ can be calculated either by finite difference:

$$\dot{P}_r(t) \equiv \frac{P_r(t + \Delta t) - P_r(t)}{\Delta t} \quad (\text{E4})$$

or by using the equations of motion, $\dot{\rho} = \mathcal{L}[\rho]$:

$$\dot{P}_r(t) = \text{Tr} \left[\hat{P}_r \dot{\rho}(t) \right] = \text{Tr} \left[\hat{P}_r \mathcal{L}[\rho(t)] \right] = \text{Tr} \left[\mathcal{L}^\dagger[\hat{P}_r] \rho(t) \right] \quad (\text{E5})$$

Where $\mathcal{L}^\dagger$ is the adjoint of the super-operator $\mathcal{L}$.

Furthermore, for symmetric potential energy surfaces, it is straight forward to show that the equilibrium distribution is uniform $P_r^{eq} = P_p^{eq} = \frac{1}{2}$. Consider e.g. the joint system-bath parity operation $\hat{S} : \begin{pmatrix} R \\ B \end{pmatrix} \rightarrow \begin{pmatrix} -R \\ -B \end{pmatrix}$, which leaves $H = H_R + \hat{R}\hat{B} + H_B$ invariant. Then:

$$\text{Tr} \left[e^{-\beta \hat{H}} \hat{P}_r \right] = \text{Tr} \left[\hat{S} e^{-\beta \hat{H}} \hat{S} \hat{P}_r \right] = \text{Tr} \left[e^{-\beta \hat{H}} \hat{S} \hat{P}_r \hat{S} \right] = \text{Tr} \left[e^{-\beta \hat{H}} \left(\mathbb{1} - \hat{P}_r \right) \right] \quad (\text{E6})$$

However, for general potential energy surfaces, it is helpful to consider alternatives to Eq. E3 for retrieving the forward reaction rate. One approach is to fit the solution of the rate equation to the expectation value of the simulation:

$$P_r(t) = \frac{k_b}{k + k_b} + \left(P_r(t_0) - \frac{k_b}{k + k_b} \right) e^{-(k+k_b)(t-t_0)} \quad (\text{E7})$$

where t_0 is typically chosen to be at a later time than the initialization of the simulation ($t = 0$) past the transients. Another approach is to consider a linear fit of Eq. E2 in terms of k and k_b e.g. through matrix calculus:

$$\begin{pmatrix} \dot{P}_r(t_0) \\ \dot{P}_r(t_1) \\ \dot{P}_r(t_2) \\ \vdots \end{pmatrix} = \begin{pmatrix} -P_r(t_0) & 1 - P_r(t_0) \\ -P_r(t_1) & 1 - P_r(t_1) \\ -P_r(t_2) & 1 - P_r(t_2) \\ \vdots & \vdots \end{pmatrix} \begin{pmatrix} k \\ k_b \end{pmatrix} \quad (\text{E8})$$

$$\Rightarrow \begin{pmatrix} k \\ k_b \end{pmatrix} = \begin{pmatrix} -P_r(t_0) & 1 - P_r(t_0) \\ -P_r(t_1) & 1 - P_r(t_1) \\ -P_r(t_2) & 1 - P_r(t_2) \\ \vdots & \vdots \end{pmatrix}^{-1} \begin{pmatrix} \dot{P}_r(t_0) \\ \dot{P}_r(t_1) \\ \dot{P}_r(t_2) \\ \vdots \end{pmatrix} \quad (\text{E9})$$

However, the transients in $P_r(t)$ can become long lived in $\dot{P}_r(t)$ due to the time-derivative increasing the amplitude of the fluctuations. Therefore, it can be advantageous to "average out" the transients by considering the integral of Eq. E2 before performing the linear fit of k and k_b .

$$\begin{pmatrix} k \\ k_b \end{pmatrix} = \begin{pmatrix} -\int_{t_0}^{t_1} P_r(t) dt & \int_{t_0}^{t_1} (1 - P_r(t)) dt \\ -\int_{t_0}^{t_2} P_r(t) dt & \int_{t_0}^{t_2} (1 - P_r(t)) dt \\ -\int_{t_0}^{t_3} P_r(t) dt & \int_{t_0}^{t_3} (1 - P_r(t)) dt \\ \vdots & \vdots \end{pmatrix}^{-1} \begin{pmatrix} P_r(t_1) - P_r(t_0) \\ P_r(t_2) - P_r(t_0) \\ P_r(t_3) - P_r(t_0) \\ \vdots \end{pmatrix} \quad (\text{E10})$$

where the integrals can be performed as a Riemann sum. By simulating for long enough times and calculating the reaction rates for different t_0 , we make sure that k has converged upon a stationary value. By comparing the different approaches, we found Eq. E10 to be the most robust with fastest convergence, in the general (asymmetric potential) setting.

1. Triple well reaction rates

In this setting, we consider the left and right well for distinct product regions with partial concentrations P_l and P_r , respectively. The middle well is the reactant region denoted by P_0 . Similar to the above, these are defined as expectation values of:

$$\hat{P}_l = \int_{-\infty}^{x_0^l} dx |x\rangle \langle x| \quad (\text{E11})$$

$$\hat{P}_0 = \int_{-x_0^l}^{x_0^r} dx |x\rangle \langle x| \quad (\text{E12})$$

$$\hat{P}_r = \int_{x_0^r}^{\infty} dx |x\rangle \langle x| \quad (\text{E13})$$

for which the rate equation becomes:

$$\begin{pmatrix} \dot{P}_l(t) \\ \dot{P}_0(t) \\ \dot{P}_r(t) \end{pmatrix} = \begin{pmatrix} P_0(t) & 0 & -P_l(t) & 0 \\ -P_0(t) & -P_0(t) & P_l(t) & P_r(t) \\ 0 & P_0(t) & 0 & -P_r(t) \end{pmatrix} \begin{pmatrix} k_{0 \rightarrow l} \\ k_{0 \rightarrow r} \\ k_{l \rightarrow 0} \\ k_{r \rightarrow 0} \end{pmatrix} \quad (\text{E14})$$

Similar to Eq. E10, the reaction rates are calculated by forming a matrix from the discrete time-series and solving the linear problem.

Supplemental Material F: Probability Currents: Transition Rate Between States

We are interested in how the rate of change in the population of the different system states depend on the environment. Using Eq. 18, we can consider the time derivative of $\hat{P}_n = |n\rangle\langle n|$, in terms of the reduced system density matrix $\rho_S(t) = \langle \mathbf{0} | \rho(t) | \mathbf{0} \rangle$. The instantaneous rate of change of $\rho_S(t)$ is:

$$\dot{\rho}_S(t) = -i \left[\hat{H}_S + \sum_k \lambda_k \hat{S}^2, \rho_S(t) \right] - i \sum_k \left[\hat{S}, \sqrt{d_k} \rho_k(t) + \sqrt{d_k^*} \rho_k^\dagger(t) \right] + \sum_i L_i \rho_S(t) L_i^\dagger - \frac{1}{2} \left\{ L_i^\dagger L_i, \rho_S(t) \right\} \quad (\text{F1})$$

with $\lambda_k = -\mathcal{I}m \left(\frac{d_k}{z_k} \right)$ and $\rho_k(t) = \langle \mathbf{0} | \hat{a}_k \rho(t) | \mathbf{0} \rangle$ ⁸⁵ and where we accommodate multiple markovian correctional terms L_i , which will be useful when we later subdivide the noise spectrum into distinct components. The Heisenberg equation of motion for $\hat{P}_n$ is then:

$$\begin{aligned} \dot{P}_n(t) = & i \text{Tr} \left[[\hat{H}_S, \hat{P}_n] \rho_S(t) \right] + i \sum_k \left(\lambda_k \text{Tr} \left[[\hat{S}^2, \hat{P}_n] \rho_S(t) \right] + \text{Tr} \left[[\hat{S}, \hat{P}_n] \left(\sqrt{d_k} \rho_k(t) + \sqrt{d_k^*} \rho_k^\dagger(t) \right) \right] \right) \\ & + \sum_i \left(\text{Tr} \left[L_i^\dagger \hat{P}_n L_i \rho_S(t) \right] - \frac{1}{2} \text{Tr} \left[\{ L_i^\dagger L_i, \hat{P}_n \} \rho_S(t) \right] \right) \end{aligned} \quad (\text{F2})$$

Utilizing references^{40,41} one can imbue the interpretation of a probability current flowing from the state n to m :

$$\dot{P}_n(t) = - \sum_m j_{n \rightarrow m}(t) \quad (\text{F3})$$

We will expand on this interpretation by decomposing the probability current further. Specifically, we will assign the different terms in Eq. F2 to different contributions $j_{n \rightarrow m}^{(i)}(t)$ of the probability current $j_{n \rightarrow m}(t) = \sum_i j_{n \rightarrow m}^{(i)}(t)$. To do this, we consider a grouping of the transitions in terms of those induced by the system Hamiltonian and those induced by coupling to the harmonic environment. Furthermore, the harmonic environment is sub-divided into contributions from the distinct spectral densities:

$$J(\omega) = \sum_i J^{(i)}(\omega) \quad \Rightarrow \quad C(t) = \sum_i \sum_k d_k^{(i)} e^{-z_k^{(i)} t} \quad (\text{F4})$$

where the $d_k^{(i)}$ and $z_k^{(i)}$ parameters are obtained by decomposing the i 'th noise spectrum independently from the others. In our case of Eq. 12, we have two distinct spectral densities such that we define three different contributions to the probability currents (Eq. 15). We denote the probability current due to H_S as *Tunneling*, the contributions from the Drude-Lorentz spectral density of the phonons as *Solvent* and the contributions associated with the spectral

density of the lossy cavity is simply referred to as *Cavity*. The probability currents are then defined as follows:

$$\partial_t \langle n | \rho_S(t) | n \rangle = - \sum_m j_{n \rightarrow m}(t) \quad (\text{F5})$$

$$j_{n \rightarrow m}(t) = j_{n \rightarrow m}^{\text{Tunneling}}(t) + j_{n \rightarrow m}^{\text{Solvent}}(t) + j_{n \rightarrow m}^{\text{Cavity}}(t) \quad (\text{F6})$$

$$j_{n \rightarrow m}^{\text{Tunneling}}(t) = 2 \text{Im} (H_{mn}^S \rho_{nm}^S(t)) \quad (\text{F7})$$

$$j_{n \rightarrow m}^{\text{Solvent}}(t) = \sum_{k \in \text{solvent}} 2 \text{Im} \left(S_{mn} 2 \text{Re}(\sqrt{d_k} \rho_{nm}^k(t)) \right) + \lambda_k 2 \text{Im} ((S^2)_{mn} \rho_{nm}^S(t)) \quad (\text{F8})$$

$$+ \frac{1}{2} ((L_s \rho_S(t))_{mn} (L_s^\dagger)_{nm} + (L_s)_{mn} (\rho_S(t) L_s^\dagger)_{nm}) \quad (\text{F9})$$

$$- \frac{1}{2} ((\rho_S(t) L_s^\dagger)_{mn} (L_s)_{nm} + (L_s^\dagger)_{mn} (L_s \rho_S(t))_{nm}) \quad (\text{F10})$$

$$j_{n \rightarrow m}^{\text{Cavity}}(t) = \sum_{k \in \text{cavity}} 2 \text{Im} \left(S_{mn} 2 \text{Re}(\sqrt{d_k} \rho_{nm}^k(t)) \right) + \lambda_k 2 \text{Im} ((S^2)_{mn} \rho_{nm}^S(t)) \quad (\text{F11})$$

$$+ \frac{1}{2} ((L_c \rho_S(t))_{mn} (L_c^\dagger)_{nm} + (L_c)_{mn} (\rho_S(t) L_c^\dagger)_{nm}) \quad (\text{F12})$$

$$- \frac{1}{2} ((\rho_S(t) L_c^\dagger)_{mn} (L_c)_{nm} + (L_c^\dagger)_{mn} (L_c \rho_S(t))_{nm}) \quad (\text{F13})$$

$$(\text{F14})$$

where L_s and L_c refer to the jump-operator for the Markovian correction (sec. D 2) of the solvent degrees of freedom and lossy cavity degrees of freedom, respectively.

In the main text, when we refer to the value of a probability current, it is the value of $j_{n \rightarrow m}(t)$ at $t = 6$ ps. At this time, the transients have died out and $j_{n \rightarrow m}(t)$ has become largely time-independent. However, $j_{n \rightarrow m}(t)$ still depends on the populations and coherences of $\rho(t)$, so it slowly decays to zero until the equilibrium distribution is reached.

1. Relating probability-currents to reaction rates.

At early times, past the transients but before the population in the reactant well has decayed substantially, the reaction rate (Eq. E3) is approximated by:

$$k \approx -\dot{P}_r(t) \quad (\text{F15})$$

We can relate this basis-independent expectation value to the basis-dependent probability currents, because we have used the localized well states in the visualization and analysis of the kinetics. Most of the spectral weight is contained to these states during the dynamics and as such:

$$P_r(t) = \text{Tr} \left(\hat{P}_r \rho(t) \right) \approx \sum_n^{\text{reactant}} \rho_{nn}(t) \quad (\text{F16})$$

where the sum is only over the states in the reactant well. The approximation stems from the fact that $\hat{P}_r$ is not diagonal in the chosen basis for $\rho(t)$. However, as long as we use these localized states, we only neglect a very small contribution and the approximation remains good. Also, crucially, n index the same basis as in $\dot{\rho}_{nn} = -\sum_m j_{n \rightarrow m}(t)$ (Eq. 14) such that we can write:

$$k \approx \sum_n^{\text{reactant}} \sum_m j_{n \rightarrow m}(t) = \sum_n^{\text{reactant product}} \sum_m j_{n \rightarrow m}(t) \quad (\text{F17})$$

where the last equality follows from the fact that $j_{n \rightarrow m}$ is anti-symmetric, i.e. the intra-reactant contributions cancel. These arguments can be extended to the case of multiple pathways. Thus, as long as the probability currents are calculated for states that are localized inside the distinct reactant and product regions and the currents are evaluated at a time t where the denominator in Eq. E3 is negligible, the reaction rates are well approximated by the sum of the currents that go across a dividing surface.

2. FGR as perturbative limit of probability currents

Here, we will gain some intuition about the probability currents introduced in sec. F by considering the Born-Markov-Secular approximation and relating the resulting expression to the well known Fermi's Golden Rule. We will begin by considering a generic system-bath setup:

$$H = H_S + \hat{S}\hat{B} + H_B \quad (\text{F18})$$

Since we consider the perturbative limit we have rightfully neglected the reorganization term in H_S . The resulting bath contribution to the probability current will be:

$$j_{n \rightarrow m}^B(t) = 2 \text{Im} \left(S_{mn} \langle n | \text{Tr}_B(\hat{B}\rho(t)) | m \rangle \right) \quad (\text{F19})$$

With $\rho(t)$ the full system plus bath density matrix. Thus, we seek to approximate $\text{Tr}_B(\hat{B}\rho(t))$. In the interaction picture $\rho_I(t) = e^{i(H_S+H_B)t}\rho(t)e^{-i(H_S+H_B)t}$ we may write:

$$\rho_I(t) = \rho_I(t_0) - i \int_{t_0}^t dt' \left[\hat{S}(t')\hat{B}(t'), \rho_I(t') \right] \quad (\text{F20})$$

Then, under the Born-Markov approximation $\rho_I(t') \approx \rho_I^S(t') \otimes \rho_B$ with $\rho_B \propto e^{-\beta H_B}$:

$$\text{Tr}_B(\hat{B}\rho(t)) \approx -i \int_{t_0}^t e^{-iH_S(t-t')} \hat{S} e^{iH_S(t-t')} \rho_S(t) C(t-t') dt' + H.C. \quad (\text{F21})$$

where we've used $\text{Tr}_B(\hat{B}(t)\rho_B) = 0$ and $C(t-t') = \text{Tr}_B(\hat{B}(t)\hat{B}(t')\rho_B)$. We can then write the probability current as:

$$j_{n \rightarrow m}^B(t) \approx 2 \text{Re} \left[- \int_0^{t-t_0} C_S^{nm}(t, \tau) C(\tau) d\tau + \int_0^{t-t_0} \left(C_S^{mn}(t, \tau) C(\tau) \right)^* d\tau \right] \quad (\text{F22})$$

$$C_S^{nm}(t, \tau) = \langle m | \hat{S} | n \rangle \langle n | e^{-iH_S\tau} \hat{S} e^{iH_S\tau} \rho_S(t) | m \rangle \quad (\text{F23})$$

When $|n\rangle$ and $|m\rangle$ are eigenstates of H_S , we get:

$$\langle n | e^{-iH_S(t-t')} \hat{S} e^{iH_S(t-t')} \rho_S(t) | m \rangle = \sum_l e^{-i(E_n - E_l)(t-t')} S_{nl} \rho_{lm}^S(t) \quad (\text{F24})$$

and we can invoke the secular approximation to argue that only the diagonal part of the density matrix contributes to the integral, such that $\sum_l S_{nl} \rho_{lm}^S(t) \approx S_{nm} \rho_{mm}^S(t)$. If we furthermore introduce the noise spectrum $C(\tau) = \int_{-\infty}^{\infty} \frac{d\omega}{2\pi} e^{-i\omega\tau} S_\beta(\omega)$, we recover:

$$j_{n \rightarrow m}^B(t) = \int_{-\infty}^{\infty} S_\beta(\omega) [\mathcal{A}^{mn}(\omega, t) - \mathcal{A}^{nm}(\omega, t)] d\omega \quad (\text{F25})$$

$$\mathcal{A}^{nm}(\omega, t) = \frac{1}{\pi} \text{Re} \left[\int_0^{t-t_0} C_S^{nm}(t, \tau) e^{-i\omega\tau} d\tau \right] \quad (\text{F26})$$

$$= |\langle m | \hat{S} | n \rangle|^2 \rho_{mm}^S(t) \times \underbrace{\frac{1}{\pi} \text{Re} \left[\int_0^{t-t_0} e^{-i(E_n - E_m)\tau} e^{-i\omega\tau} d\tau \right]}_{\int_{-(t-t_0)}^{t-t_0} \frac{d\tau}{2\pi} e^{-i(\omega_{nm} + \omega)\tau}} \quad (\text{F27})$$

$$\quad (\text{F28})$$

such that in the long time limit $t \gg t_0$:

$$j_{n \rightarrow m}^B \approx |S_{nm}|^2 S_\beta(\omega_{nm}) \rho_{nn}^S - |S_{nm}|^2 S_\beta(-\omega_{nm}) \rho_{mm}^S \quad (\text{F29})$$

That is, the probability current from n to m becomes the population in n times the FGR rate from n to m minus the population in m times the FGR rate from m to n (the two rates are related by $S_\beta(-\omega) = e^{-\beta\omega} S_\beta(\omega)$, see sec. A). Thus, when the probability current is subject to the same approximations as FGR, the resulting expression is consistent with FGR.

a. Broadening of transition frequencies due to phonon bath

The derivation above (and FGR in general) relied on $|n\rangle$ and $|m\rangle$ being eigenstates of H_S . However, in the present case, we are interested in intra-well transitions between localized states, which are not eigenstates of H_S . The reason that FGR can still be applied as is, is because a secondary bath, in this case the solvent, renders these localized states metastable by suppressing the tunneling between the constituent eigenstates (this can be seen in Fig. 13 where the bath-dressed states look like the localized states in Fig. 7a, even though no such post-processing localization took place when visualizing the states in Fig. 13). Furthermore, the secondary bath broadens the transition frequencies of the system by inducing a finite lifetime of constituent states.

To incorporate these ideas into the result above, we may view H_S as an enlarged system that not only contains the system degrees of freedom, but now also incorporates e.g. the phonon degrees of freedom. In order to recover a succinct expression for $C_S^{nm}(t, \tau)$ for this enlarged system, we make the following considerations. The Markov approximation simply says that, when evaluating the integral F21, we may consider the dynamics of $\rho_S(t - \tau) \approx e^{iH_S\tau}\rho_S(t)e^{-iH_S\tau}$ to be closed under H_S (i.e. we disregard the perturbing bath when calculating the induced transition rates). In the current case where H_S contains the phonon bath, the dynamics are not only closed but it will evolve towards the thermal equilibrium distribution of the system plus phonon bath ρ_S^{eq} . Such that in the long time limit and within this approximation $C_S^{nm}(\tau) \approx \text{Tr} \left[e^{iH_S\tau} |m\rangle \langle m| \hat{S} |n\rangle \langle n| e^{-iH_S\tau} \rho_S^{eq} \right]$ which resembles the dipole-dipole correlation function $C_S(\tau)$, except it is restricted to the pathway between n and m : $C_S(\tau) = \text{Tr} \left[e^{iH_S\tau} \hat{S} e^{-iH_S\tau} \hat{S} \rho_S^{eq} \right] = \sum_{nm} C_S^{nm}(\tau)$. Thus, the simplest way to incorporate dissipation, is to broaden the delta function in Eq. F27 by including decay:

$$\mathcal{A}^{nm}(\omega, t) = |\langle m|S|n\rangle|^2 \rho_{mm}^S(t) \underbrace{\text{Re} \int_0^\infty \frac{d\tau}{\pi} e^{-i(\omega_{nm} + \omega)\tau - \Gamma\tau}}_{\frac{1}{\pi} \frac{\Gamma}{(\omega_{nm} + \omega)^2 + \Gamma^2}} \quad (\text{F30})$$

where the inverse lifetime Γ and the possibly renormalized transition frequency ω_{nm} , can be read-off from the part of $C_S(\tau)$ that is due to the $n \rightarrow m$ transition. In our example, Fig. 9, we did not attempt to find Γ and ω_{nm} a priori, but simply found $\Gamma = 30\text{cm}^{-1}$ and $|\omega_{nm}| = 1205\text{cm}^{-1}$ to best explain the results across all θ and ω_c . The value for Γ align well with what has earlier been found for this system⁷². However, the bare transition frequency is $\approx 1195\text{cm}^{-1}$ indicating that the phonon bath not only broadens the transition but also shifts it slightly.

Supplemental Material G: Distribution of reaction rates and comparison to isotropic averaging

For the symmetric double well PES, we can write the difference of the population in the two wells (Eq. E7) as $\Delta P(t) = \Delta P_0 e^{-2kt}$. Based on the agreement seen in Fig. 9, we can use the FGR rate in Eq. 27 to study what the reaction rate profile would look like for a given molecule orientation, as parameterized by ϕ . We can then imagine an ensemble of molecules with random orientations and since the orientation affects the reaction, this yields a distribution of reaction rates⁸⁶. Thus, instead of the average reaction rate $\langle k(\phi) \rangle_\phi$, which was studied in Fig. 9, the primary experimentally relevant entity is:

$$\Delta P(t) = \langle \Delta P(t, \phi) \rangle_\phi = \Delta P_0 \int_0^\pi p(\phi) e^{-2k(\phi)t} d\phi \quad (\text{G1})$$

where $p(\phi)$ is the distribution of the angle between the molecule dipole moment and the cavity polarization. E.g. $p(\phi) = \frac{1}{\pi}$ as in sec. IV B or $p(\phi) = \frac{1}{2} \sin(\phi)$ for a uniform distribution of orientations in three dimensional space. The distribution of reaction rates $p(k)$ can then be defined by a change of variables:

$$\Delta P(t) = \Delta P_0 \int_{k_{\min}}^{k_{\max}} p(k) e^{-2kt} dk \quad (\text{G2})$$

where $p(k) \propto p(\phi) \frac{d\phi}{dk}$. As an example, consider the harmonic environment without the spectator mode (Eq. 12). Using Eq. 27, as was done in²⁰, we schematically recover: $k(\phi) = k_0 + \cos^2(\phi)k_1$, where k_0 is the reaction rate outside the cavity and k_1 is the maximal contribution from the cavity. Using the isotropic average, one recovers the following distribution of reaction rates: $p(k) = \frac{1}{2\sqrt{k_1(k-k_0)}}$ which naturally reproduces the average reaction rate:

$$\int_{k_0}^{k_0+k_1} k p(k) dk = k_0 + \frac{1}{3} k_1 = \langle k(\phi) \rangle_\phi$$

In this kind of simplified scenario, where we consider an ensemble of molecules whose reaction rate is described by the distribution $p(k)$, we could imagine an observer fitting the data $\Delta P(t)$ with a single exponential. The observer would find a rate k_{fit} that lies between the result at early times $\langle k \rangle$ and the result obtained from the long time asymptotics $\langle k^{-1} \rangle^{-1}$. This kind of distribution of rates can perhaps be seen in the experimental data presented in Fig. 2E of ref.⁵. While the errorbars make it difficult to make a precise comparison, the mean-values suggests that there is an initial fast decay and a subsequent slower decay and that the fit will yield a value that lies in between these two rates.

Now, for the example in sec. IV B, we would like to understand what the distribution of rates look like for both angular distributions $p(\phi) = \frac{1}{\pi}$ and $p(\phi) = \frac{1}{2} \sin(\phi)$. Figure 12 show these distributions alongside the resulting fast rate $\langle k \rangle$ and the slow rate $\langle k^{-1} \rangle^{-1}$. We see that these two rates are comparable, so, in this example, $\langle k \rangle$ is representative of what one could expect to obtain from a fit to an ensemble of molecules with random orientations. Furthermore, the angular distributions lead to qualitatively similar results, so even though $p(\phi) = \frac{1}{\pi}$ was used in sec. IV B, the discussion there also applies to the scenario where the molecule orientations are isotropically distributed.

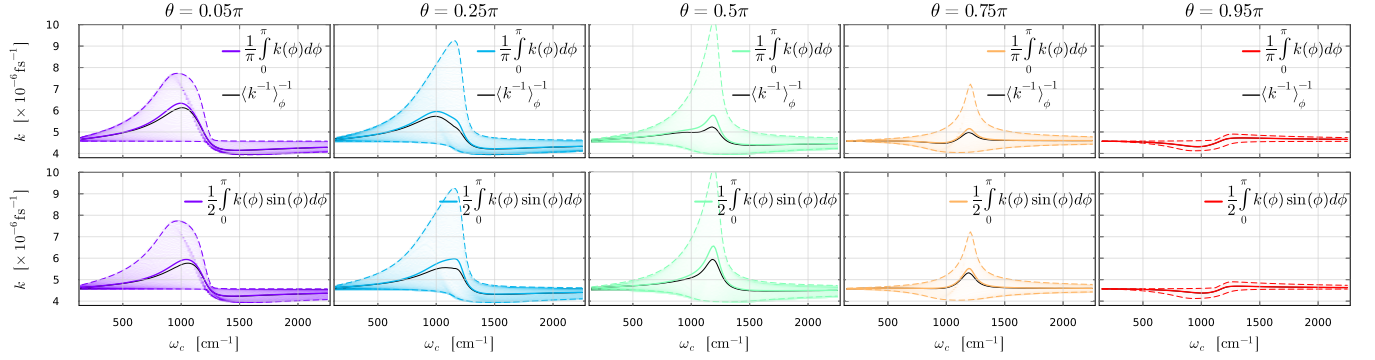


FIG. 12. Distribution of reaction rates $p(k)$ for an ensemble of molecules with random orientations. The rate is calculated from Eq. 27 and the distribution of the molecule orientations is either uniform $p(\phi) = \frac{1}{\pi}$ (top panel) or isotropic $p(\phi) = \frac{1}{2} \sin \phi$ (lower panel). The dashed lines represent the minimum and maximum of $p(k)$ and the full colored line is the mean $\langle k \rangle$ while the black line is the asymptotic reaction rate $\langle k^{-1} \rangle^{-1} = (\int_0^\infty p(k) k^{-1} dk)^{-1}$. Fitting a single exponential to the dynamics of such an ensemble, would yield a reaction rate in between these two limits.

Supplemental Material H: Equilibrium state of the reaction coordinate

For an N-level system coupled to a dissipative environment, such as the one considered in this work and under fairly general conditions³⁹, it has been proven that the long-time asymptotics of system observables $\mathcal{O}_S$ is given by:

$$\lim_{t \rightarrow \infty} \text{Tr} \left[e^{-i(H_S + H_{SB} + H_B)t} \rho_S(0) \otimes \frac{e^{-\beta H_B}}{\text{Tr} e^{-\beta H_B}} e^{i(H_S + H_{SB} + H_B)t} \mathcal{O}_S \right] = \text{Tr}_S [\tau_{\text{MF}} \mathcal{O}_S] \quad (\text{H1})$$

where τ_{MF} is the *mean force Gibbs state*:

$$\tau_{\text{MF}} = \text{Tr}_B \left[\frac{e^{-\beta(H_S + H_{SB} + H_B)}}{\text{Tr} [e^{-\beta(H_S + H_{SB} + H_B)}]} \right] \quad (\text{H2})$$

While proving the same for infinite-dimensional systems with a discrete spectra seems to remain an open problem, it has been established for the damped harmonic oscillator. Since we have utilized a finite dimensional representation of our system in the dynamical simulation, the above result will be able to explain our observations. The mean force Gibbs state has been calculated perturbatively in the bath coupling⁶⁶:

$$\tau_{\text{MF}} = \tau + \tau_{\text{MF}}^{(2)} + \dots \quad (\text{H3})$$

where $\tau \propto e^{-\beta H_S}$ is the system Gibbs state and $\tau_{\text{MF}}^{(2)}$ is the leading correction. Whether this perturbative expansion applies is governed by:

$$\lambda := \sqrt{\left| \beta \sum_{nm} |S_{nm}|^2 \frac{e^{-\beta E_n}}{\sum_l e^{-\beta E_l}} D_\beta(E_n - E_m) \right|} \ll 1 \quad (\text{H4})$$

where $\hat{S}$ is the system operator that couples to the bath and the function $D_\beta(\nu)$, which is also used in the expression for $\tau_{\text{MF}}^{(2)}$, is:

$$D_\beta(\nu) = \mathcal{P} \int_0^\infty d\omega J(\omega) \left(\frac{\nu \coth(\beta\omega/2) + \omega}{\omega^2 - \nu^2} - \frac{1}{\omega} \right) = \frac{\nu}{2} \mathcal{P} \int_{-\infty}^\infty d\omega \frac{S_\beta(\omega)}{\omega(\omega - \nu)} \quad (\text{H5})$$

which for the decomposed noise spectra (Eq. 22) reduces to:

$$D_\beta(\nu) = \pi\nu \sum_k \text{Im} \left[\frac{d_k}{z_k(\nu + iz_k)} \right] \quad (\text{H6})$$

For the systems and noise spectra considered in this study $\lambda \sim 0.195 - 0.227$. We will calculate ρ_{MF} to second order using Eq. 5 while neglecting terms beyond fourth order $\lambda^4 \sim 5\%$. If we want to consider this kind of perturbative methodology for stronger coupling, the method known as *reaction coordinate mapping* (see ref.⁵³) could prove useful to reduce the perturbative error. The basic idea is to include strongly coupled degrees of freedom in H_S and perturbatively expand in the residual (and hopefully) small couplings.

We evaluate the population inside each well $P_{\text{left/middle/right}}^{\text{eq}}$ which, for the double well PES, we can compare to the reaction rates through $\frac{k_{\rightarrow}}{k_{\leftarrow}} = \frac{P_{\text{right}}^{\text{eq}}}{P_{\text{left}}^{\text{eq}}}$. Furthermore, we can visualize τ_{MF} in terms of its eigendecomposition $\tau_{\text{MF}} = UDU^\dagger$. The eigenstates U correspond to the bath dressed vibrational states in thermal equilibrium, with the frequencies given by $\omega_{nm} = -\frac{1}{\beta} (\ln D_n - \ln D_m)$.

1. Mean force Gibbs state for the triple well, Fig. 7

In Figure 13, the effective system (described above) is visualized for the triple well PES coupled to the phonon bath. Compared to Fig. 7a, we see that all the excitations have been somewhat lowered in energy. However, the left well has been favored due to a more significant lowering of the energy of its excited state. More importantly, this is also reflected in the energy of the left wells ground state which has been decreased, while the energy of the right well ground state has been increased, slightly.

An understanding that can explain these observations, is that the environment observes the position ($\hat{S} = \hat{R}$) of the reaction coordinate, which localizes the wavefunction. The reason this favors the left well is because it has lower potential energy. In other words, the kinetic energy is anyway increased by interacting with the environment, so it might as well save some potential energy by residing more in the left well.

We quantify this effect in the second part of figure 13, where we also include the effect of the cavity on the equilibrium population of each well. We see a sizeable correction to the prediction given by the Gibbs state τ (dotted lines), due to the phonon bath. However, the presence of the cavity has nearly no effect on the equilibrium distribution. Especially, we do not see any resonant behavior and the cavity seems to only enhance the effect described above, where the localization of the reaction coordinate favors the deeper left well. Also, when we tried to consider a non-linear dipole moment surface, the results we saw were similar.

2. Mean force Gibbs state for the double wells, Fig. 5

For the double well systems, we can compare the predictions of the mean force Gibbs state to those obtained from the dynamical simulation $P_{\text{left/right}}^{\text{eq}} = \frac{k_{\leftarrow/\rightarrow}}{k_{\rightarrow/\leftarrow}}$. This is plotted in figure 14 alongside the distribution determined by the Gibbs state τ and its leading bath-induced correction $\tau_{\text{MFG}}^{(2)}$. The ω_r refers to the frequency of the right well and the ordering of the subfigures from left to right is exactly the same as in Fig. 6.

Again, we see the same behavior that the narrow well is favored because it has lower potential energy once the wavefunction is localized. The correction $\tau_{\text{MFG}}^{(2)}$ is especially noticeable for $\omega_r \approx 500\text{cm}^{-1}$ since it switches which $P_{\text{left/right}}^{\text{eq}}$ is dominant when compared to τ .

It is difficult to establish quantitative agreement to the numerical predictions via the reaction rates, because the numerical noise is magnified from the mutual division of $k_{\leftarrow}$ and $k_{\rightarrow}$. These numerical predictions display an ω_c dependence, which could be an artifact from this kind of short-time extrapolation to long-time asymptotics. For a proper comparison, one could consider obtaining the steady state of the master equation directly, Eq. 18

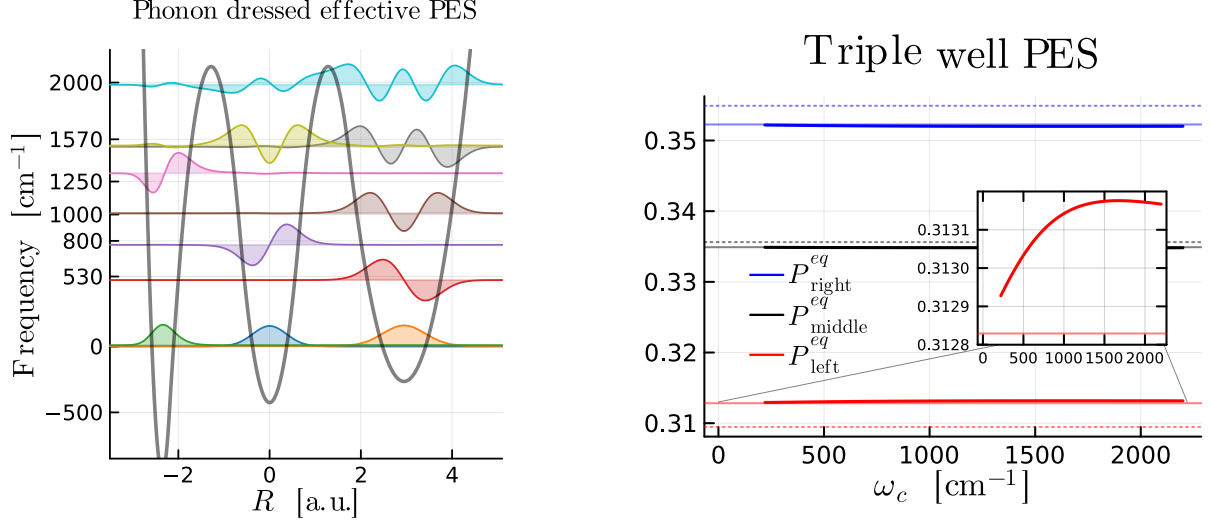


FIG. 13. (left) Visualization of the phonon-dressed vibrational states of the reaction coordinate. (right) Comparison of the population in the left (red), middle (black) and right (blue) wells at equilibrium. The dotted lines are the predictions from the Gibbs state $\tau \propto e^{-\beta H_R}$. The solid lines include the correction from the structured environment via Eq. H3, for which the horizontal line is for outside the cavity while the thicker line includes the cavity contribution.

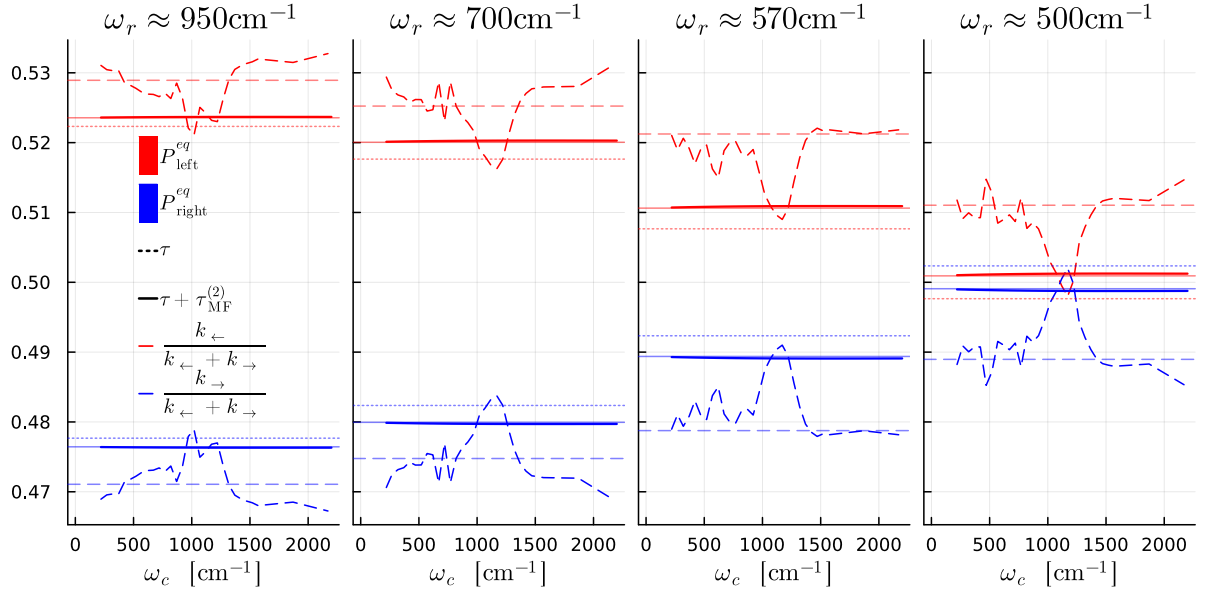


FIG. 14. Comparison of the population in the left (red) and right (blue) wells at equilibrium for the double wells in Fig. 6. The dotted lines are the predictions from the Gibbs state $\tau \propto e^{-\beta H_R}$. The solid lines include the correction from the structured environment via Eq. H3, for which the horizontal line is for outside the cavity while the thicker line includes the cavity contribution to the noise spectrum. The dashed lines represent the empirical prediction from the normalized reaction rates.

Supplemental Material I: Kinetic and thermodynamic control of the triple well system

The triple well PES gives rise to reversible parallel reactions, which can be described by the rate equation:

$$\begin{pmatrix} \dot{P}_l \\ \dot{P}_0 \\ \dot{P}_r \end{pmatrix} = \begin{pmatrix} -k_{l0} & k_{0l} & 0 \\ k_{l0} & -(k_{0l} + k_{0r}) & k_{r0} \\ 0 & k_{0r} & -k_{r0} \end{pmatrix} \begin{pmatrix} P_l \\ P_0 \\ P_r \end{pmatrix} \quad (\text{I1})$$

with $P_0(t=0) = 1$ and $k_{ij} := k_{i \rightarrow j}$, i.e. this is the same rate equation as Eq. E14. In order to be probability preserving, one of the eigenvalues of the matrix are zero. The associated eigenvector is the steady state distribution:

$$\lim_{t \rightarrow \infty} \begin{pmatrix} P_l(t) \\ P_0(t) \\ P_r(t) \end{pmatrix} \propto \begin{pmatrix} k_{0l}k_{r0} \\ k_{l0}k_{r0} \\ k_{l0}k_{0r} \end{pmatrix} \quad (I2)$$

Already here, we see why we should not expect a modification of the steady state distribution from simply manipulating the left or right reaction rates. E.g. if either k_{0l} or k_{0r} is scaled by a factor α , the reverse rate k_{l0} , k_{r0} is also scaled by α since they are linked by detailed balance. Then, every well distribution acquires the same scaling, which disappears when accounting for normalization. The most direct way to affect the equilibrium distribution would be to change the temperature which affects the detailed balance condition and thus the equilibrium populations. The regime in which the product distribution is determined by these equilibrium populations rather than by the rates of formation is referred to as thermodynamic control⁸⁷.

However, the timescales of how the distribution reaches this steady state is determined by the remaining two eigenvalues of the rate matrix:

$$-\gamma_{\pm} = \frac{K_l + K_r}{2} \pm \frac{\Delta}{2} \quad (I3)$$

$$\text{with } K_l = k_{0l} + k_{l0}, \quad K_r = k_{0r} + k_{r0} \quad (I4)$$

$$\text{and } \Delta = \sqrt{(K_l + K_r)^2 - 4(K_l K_r - k_{0l} k_{0r})} \quad (I5)$$

In the irreversible limit $k_{l0} \approx 0 \approx k_{r0}$ the two decay rates become a slow $\gamma_- \approx 0$ and a fast mode $\gamma_+ \approx k_{0l} + k_{0r}$. During the time $\mathcal{O}(\gamma_+^{-1})$ the middle well has depleted into both the left and the right well. The population in either well is determined by the branching ratio $P_l(t \sim \gamma_+^{-1}) \approx \frac{k_{0l}}{k_{0l} + k_{0r}}$, $P_r(t \sim \gamma_+^{-1}) \approx \frac{k_{0r}}{k_{0l} + k_{0r}}$ which constitutes the kinetic yield. The kinetic yield is straightforwardly affected by changes in the reaction rates. The regime in which the product distribution is governed by relative reaction rates rather than equilibrium populations, is generally referred to as kinetic control⁸⁷. Nonetheless, as long as the reaction is not completely irreversible ($k_{l0} \neq 0 \neq k_{r0}$), the distribution will eventually reach the steady state. Accounting for the non-zero backward rates, the kinetic yield becomes $P_l(t \sim \gamma_+^{-1}) \approx \frac{k_{0l}(\gamma_+ - k_{r0})}{\gamma_+(\gamma_+ - \gamma_-)}$, $P_r(t \sim \gamma_+^{-1}) \approx \frac{k_{0r}(\gamma_+ - k_{l0})}{\gamma_+(\gamma_+ - \gamma_-)}$ Figure 15 shows two examples of the dynamics of the

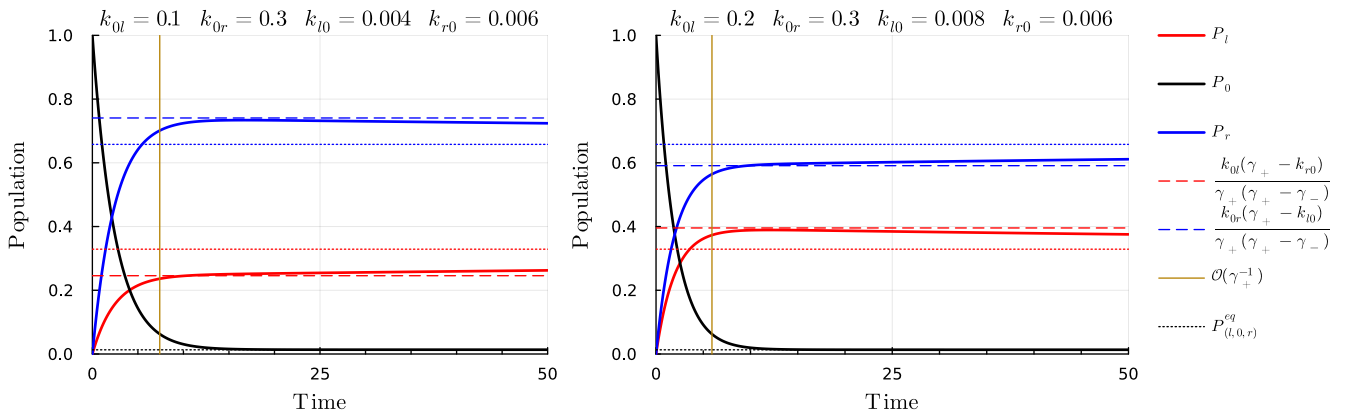


FIG. 15. Example of the population dynamics for the triple well rate equation $P_l \xrightleftharpoons[k_{l0}]{k_{0l}} P_0 \xrightleftharpoons[k_{r0}]{k_{0r}} P_r$. The dotted lines are the equilibrium distributions while the dashed lines are the kinetic yield.

population for the reversible parallel rate equation, Eq. I1. The horizontal lines represent the equilibrium distribution (dotted) and the kinetic yield (dashed). The kinetic yield best describe the distribution of the products in the intermediate time window $\mathcal{O}(\gamma_+^{-1}) \lesssim t \lesssim \mathcal{O}(\gamma_-^{-1})$ as defined in Eq. I3. The kinetic yield is affected by the doubling of the rates between the left well and the reactant well whereas the equilibrium distribution remains unaffected. In both examples, the populations slowly evolve towards the equilibrium distribution.

Supplemental Material J: The phonon bath and Kramer's turnover

The phonon spectral density is the same for all simulations. It is of the Drude Lorentz form with

$$J_R^{(DL)}(\omega) = \frac{2\lambda_R\gamma_R\omega}{\gamma_R^2 + \omega^2} \quad \lambda_R = \frac{\eta_\nu M\gamma_R}{2} \times 1000\text{cm}^{-1} \approx \eta_\nu \times 0.00381157\text{a.u.} \quad \gamma_R = 200\text{cm}^{-1} \quad (\text{J1})$$

The reaction rate for the symmetric double well coupled to the phonon bath, were calculated for different coupling strengths η_ν , as seen in Fig. 16. In the limit of vanishing η_ν the state of the system would oscillate between the

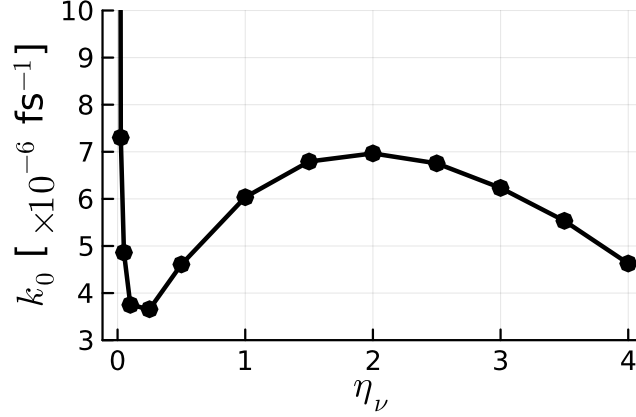


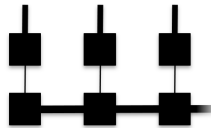
FIG. 16. Reaction rates for the symmetric double well coupled to the phonon bath for different coupling strengths η_ν . For all other simulations $\eta_\nu = 0.1$ was used.

wells on a timescale $\sim 600 \times 10^{-6}\text{fs}^{-1}$. By increasing $0 < \eta_\nu < 0.25$, the coupling to the bath greatly suppresses the tunneling. In this regime, the reaction is limited by the sparse energy transactions with the environment, so by increasing further $0.25 < \eta_\nu < 2$, the bath helps drive the transition until the crossover $\eta_\nu \approx 2$. From this point, the reaction rate decreases due to the increased localization from strongly coupling to the environment.

a. Tensor network simulations in the strong coupling regime

In obtaining the above results, we looked into different tensor network techniques to alleviate the numerical hurdles that arise in the strong coupling regime $3 \lesssim \eta_\nu$. The primary obstacles are that the cutoff of the auxiliary modes need to be increased together with the bond-dimension of the tensor network ansatz.

In an attempt to compress the local Hilbert space dimensions, we applied the Optimized Boson Basis or Local Basis Optimization approach⁸⁸. In short, the three legged tensors of the MPS, which all have high local dimension in this scenario, are decomposed such that the physical index is separated from the two virtual bond indices. This introduces an intermediate index that connects the two tensors, and the goal is to substantially compress this new bond, such that the computational cost of the MPS representation is reduced.



However, our experience is that such a compression is not achievable for this scenario, and that the bond that connects the two new tensors had to be too large to render the approach useful.

Another approach we tested is the pseudosite method⁸⁹. The bosonic occupation number is represented in binary form, such that a bosonic degree of freedom with a local cutoff of e.g. $64 = 2^6$ is mapped exactly onto six two-dimensional pseudosites. The joint state of these six pseudosites therefore encodes the Fock state of the original bosonic mode. Bosonic operators can be represented in terms of operators acting on the pseudosites, allowing the original bosonic site to be treated within a standard tensor-network framework. By distributing the original large

local Hilbert space over several two-dimensional sites, standard tensor-network compression can be applied to the resulting pseudosite representation, potentially reducing the computational cost.



Doing this for each bosonic degree of freedom, we again experienced that this did not reduce the computational cost because the resulting tensor network could not compress enough while still keeping the necessary accuracy.

In the end, we simply simulated the strong-coupling scenarios using 1-site TDVP with GPU acceleration⁹⁰. Methods such as the reaction coordinate mapping (see ref.⁵³) might also prove also for simulating the dynamics of such scenarios.

Supplemental Material K: Additional asymmetric double well probability currents

Figure 17 provides a complete overview of the probability currents for the systems shown in Fig. 5.

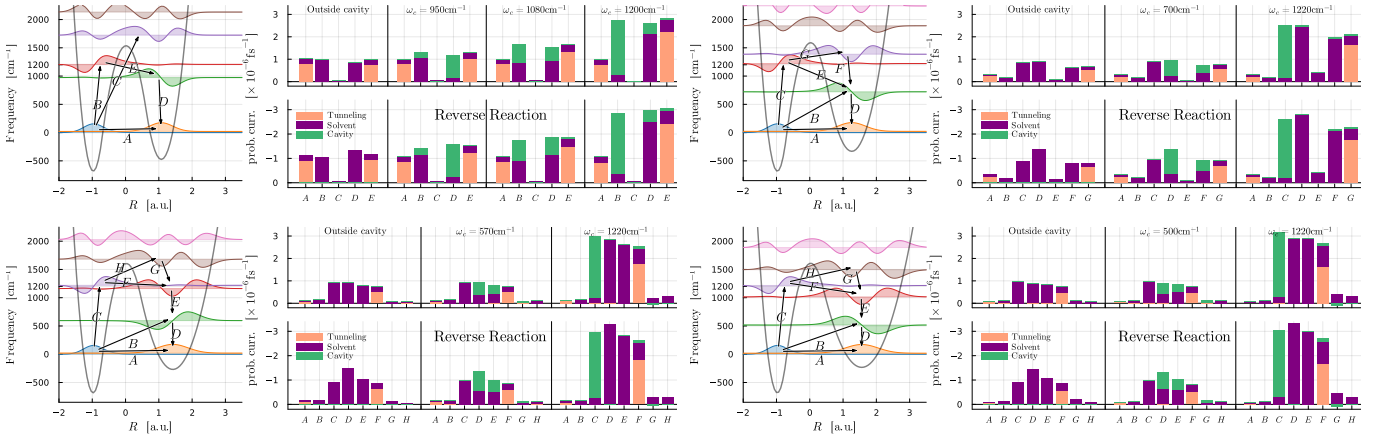


FIG. 17. Probability currents for the asymmetric double wells. A negative rate means that the current goes in the opposite direction of the arrow. All currents with magnitude larger than $0.05 \times 10^{-6} \text{fs}^{-1}$ are shown.

Supplemental Material L: Overview of parameter values

Symbol	Description	Value	Unit
Reaction Coordinate PES:			
Symmetric Double Well			
M	Effective mass	1836	a.u.
E_b	Barrier height	2120	cm^{-1}
ω_0	Barrier frequency	866.020	cm^{-1}
$\omega_l = \omega_r$	Reactant and product well frequency	1414.177	cm^{-1}
Asymmetric Double Well			
E_b	Barrier height	2120	cm^{-1}
ω_0	Barrier frequency	866.020	cm^{-1}
ω_l	Reactant well frequency	1414.177	cm^{-1}
ω_r	Product well frequency	{999.914, 707.046, 577.301, 500.077}	cm^{-1}
Triple Well			
E_{ll}	Left well offset	960.660	cm^{-1}
E_l	Left barrier height	2120	cm^{-1}
E_0	Middle well offset	426.963	cm^{-1}
E_r	Right barrier height	2120	cm^{-1}
E_{rr}	Right well offset	266.695	cm^{-1}
ω_{ll}	Left well frequency	2121.320	cm^{-1}
ω_l	Left barrier frequency	866.02	cm^{-1}
ω_0	Middle well frequency	853.927	cm^{-1}
ω_r	Right barrier frequency	866.02	cm^{-1}
ω_{rr}	Right well frequency	533.389	cm^{-1}
Harmonic Environment:			
T	Temperature	300	K
η_c	Light-matter interaction strength	0.1	—
$\tilde{\eta}_R$	Reaction coordinate - cavity coupling	1.0	—
$\tilde{\eta}_Q$	Spectator mode - cavity coupling	0.1	a.u.
C_Q	Reaction coordinate - Spectator mode coupling	2.2326×10^{-5}	a.u.
ω_Q	Spectator mode frequency	1170	cm^{-1}
Cavity loss parameters			
τ_c	Cavity lifetime	100	fs
α	Cavity Markovian parameter	$(1 - e^{-\beta\omega_c})/\tau_c$	—
Reaction coordinate Drude Lorentz bath			
γ_R	Characteristic frequency	200	cm^{-1}
λ_R	Reorganization energy	83.654	cm^{-1}
Spectator mode Drude Lorentz bath			
λ_Q	Reorganization energy	0.147	cm^{-1}
γ_Q	Characteristic frequency	6000	cm^{-1}
Bath decomposition hyper-parameters			
$[\omega_l, \omega_h]$	Frequency window	[-0.99,1]	a.u.
Tolerance	Criterion for convergence (as low as possible)	e.g. $1e-14$	—
Lawson iterations	post-processing step	20	—

TABLE I. Overview of simulation parameters.

Description	Value range
Simulation parameters:	
System Hilbert space dimension	12 to 30
Fock space cutoff	4 to 75
Maximum value of $2\Gamma_0\tau_0$ (for the Markovian compression, sec. D 2)	0.01 with typical values of $\mathcal{O}(0.001)$
bond dimension	4 to 600
Tolerance of the Krylov update for the local propagation in TDVP	1e-8
Time step	0.25fs
Also, the probability currents are $j_{n\rightarrow m}(t)$ evaluated at:	$t = 6\text{ps}$

TABLE II. Additional simulation parameters. Convergence has been carefully checked for these. The extra large Fock space cutoff and bond dimension was mainly used for the strong coupling scenarios in sec. J.